Industrial pH Meter
HP-200

Instruction Manual

CODE:GZ0000214572

Preface

This manual describes the operation of the Industrial pH Meter, HP-200.
Be sure to read this manual before using the product to ensure proper and safe operation of the instrument. Also safely store the manual so it is readily available whenever necessary.

Product specifications and appearance, as well as the contents of this manual are subject to change without notice.

Warranty and Responsibility

HORIBA Advanced Techno, CO., Ltd. warrants that the Product shall be free from defects in material and workmanship and agrees to repair or replace free of charge, at option of HORIBA Advanced Techno, CO., Ltd., any malfunctioned or damaged Product attributable to responsibility of HORIBA Advanced Techno, CO., Ltd. for a period of one (1) year from the delivery unless otherwise agreed with a written agreement. In any one of the following cases, none of the warranties set forth herein shall be extended;

- Any malfunction or damage attributable to improper operation
- Any malfunction attributable to repair or modification by any person not authorized by HORIBA Advanced Techno, CO., Ltd.
- Any malfunction or damage attributable to the use in an environment not specified in this manual
- Any malfunction or damage attributable to violation of the instructions in this manual or operations in the manner not specified in this manual
- Any malfunction or damage attributable to any cause or causes beyond the reasonable control of HORIBA Advanced Techno, CO., Ltd. such as natural disasters
- Any deterioration in appearance attributable to corrosion, rust, and so on
- Replacement of consumables

HORIBA ADVANCED TECHNO, CO., LTD. SHALL NOT BE LIABLE FOR ANY DAMAGES RESULTING FROM ANY MALFUNCTIONS OF THE PRODUCT, ANY ERASURE OF DATA, OR ANY OTHER USES OF THE PRODUCT.

Trademarks

Generally, company names and brand names are either registered trademarks or trademarks of the respective companies.

Regulations

Conformable Directive

This equipment conforms to the following directives and standards:

Directives:	the EMC Directive 2004/108/EC the Low Voltage Directive 2006/95/EC
Standards:	[the EMC Directive] EN61326-1: 2006 Class A, industry [the Low Voltage Directive] EN61010-1: 2001

Note

When the sensor cable, transmission cable, or relay cable is extended to 30 meters or more, the surge test in the EMC Directive for CE Marking is not applicable.

Installation Environment

This product is designed for the following environment.

- Installation Categories II
- Pollution degree 2

FCC Rules

Any changes or modifications not expressly approved by the party responsible for compliance shall void the user's authority to operate the equipment.

WARNING

This equipment has been tested and found to comply with the limits for a Class A digital device, pursuant to part 15 of the FCC Rules. These limits are designed to provide reasonable protection against harmful interference when the equipment is operated in a commercial environment. This equipment generates, uses, and can radiate radio frequency energy and, if not installed and used in accordance with the instruction manual, may cause harmful interference to radio communications.
Operation of this equipment in a residential area is likely to cause harmful interference in which case the user will be required to correct the interference at his own expense.

For Your Safety

Hazard Classification and Warning Symbols

Warning messages are described in the following manner. Read the messages and follow the instructions carefully.

- **Hazard classification**

⚠DANGER	This indicates an imminently hazardous situation which, if not avoided, will result in death or serious injury. This signal word is to be limited to the most extreme situations.
⚠WARNING	This indicates a potentially hazardous situation which, if not avoided, could result in death or serious injury.
⚠CAUTION	This indicates a potentially hazardous situation which, if not avoided, may result in minor or moderate injury. It may also be used to alert against unsafe practices. Without safety alert indication of hazardous situation which, if not avoided, could result in property damage.

- **Warning symbols**

Description of what should be done, or what should be followed

Description of what should never be done, or what is prohibited

Safety Precautions

This section provides precautions to enable you to use the product safely and correctly and to prevent injury and damage. The terms of DANGER, WARNING, and CAUTION indicate the degree of imminency and hazardous situation. Read the precautions carefully as it contains important safety messages.

⚠ WARNING
When any load larger than the contact capacity or any induction load (e.g., a motor or a pump) is used, be sure to connect that load via a power relay that has a rating larger than the load rating.
When the HP-200 may be subject to lightning, install a conductor rod on the output side of the HP-200 and on the side of receiving instruments.
Be sure to connect the grounding terminal (class D grounding) for safety.
Electric shock Be sure to check that no power is supplied before starting work.
Strong acid If dilute hydrochloric acid gets into the eye, mucous membrane may be irritated, resulting in the loss of sight. In handling hydrochloric acid, be sure to use protective goggles, protective gloves, and a protective mask. If hydrochloric acid gets into the eye, immediately rinse the eye with a large quantity of water for at least 15 minutes and then receive medical attention. (When rinsing the eye, well open it with fingers so that the eyeball and the eyelid can be fully cleaned.) If hydrochloric acid adheres to your body or clothes, you may get burned. Therefore, immediately remove the clothes and then rinse off hydrochloric acid with a large quantity of water.

⚠ CAUTION
If there is corrosive gas in the installation environment, send clean instrumentation air from the air purge inlet on the HP-200.
When the HP-200 is OFF, the C-NC contact is short-circuited. Therefore, be careful about the connection of any load.
Since operation of the HP-200 at any voltage outside the rated range can cause a fault, check the power supply voltage. Also check that the voltage fluctuation range of the power source falls within the range of rated voltage ±10%.
The electrode is a glass product, which is broken if it is exposed to an impact, strong force, or the like. In handling the electrode, therefore, exercise sufficient care.

Product Handling Information

Operational Precautions

Use of the equipment in a manner not specified by the manufacturer may impair the protection provided by the equipment. And it may also reduce equipment performance.
Therefore, exercise the following precautions:

- Do not press any operation key with the tip of your nail or any pointed thing.
- Do not use organic solvent or the like.
- Do not immerse the HP-200 in dilute hydrochloric acid for long hours.

Disposal of the Product

When disposing of the product, follow the related laws and/or regulations of your country for disposal of the product.

Manual Information

Description in This Manual

Note

This interprets the necessary points for correct operation and notifies the important points for handling the unit.

Reference

This interprets the necessary points for correct operation and notifies the important points for handling the unit.

Tip

This indicates the part of where to refer the information.

Definitions of terms

This document uses the following terms as defined:

Term	Definition
Hold down	Maintain pressing a key until the indicator turns ON or the display changes.
Press	Lightly press a key once.
Flash	Quickly flash several times, indicating that the setting has been established.

Contents

Overview

Introduction

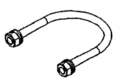
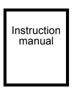
Thank you very much for purchasing our model HP-200 pH Meter for industrial use.
The HP-200 allows you to make a measurement when the pH electrode and a power source for 100 V to 240 VAC are connected.
The measured value and various parameters are displayed on the LCD part. When used with our cleaner, the HP-200 allows you to control the cleaner.
A variety of self-diagnostic capabilities is provided to allow you to detect a trouble with the pH electrode or the HP-200.

● Features

- Outdoor installation type (equivalent to IP65; splash-proof construction)
- Selectable simultaneous display of temperature
- All settings available with front keys
- Applicable for 5 kinds of standard solutions (pH 7 plus one to three among pH 2, 4, 9, and 10)
- Improved maintenance feature (self-diagnostic capability)
- Selectable transmission output range
- Backup of stored data
- Easy-to-read display (150% larger than former display)
- Improved operability of keys by using an emboss sheet
- Improved mode display by using icons
- 4 kinds of temperature compensation electrodes (500, 6.8 k, 1 k, and 10 k) self-detection capability provided

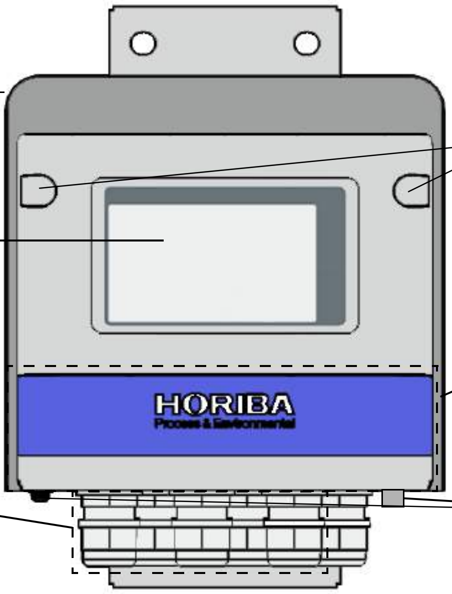
Content

● Items included

Name	Illustration	Quantity	Remarks
Main body		1 set	With bracket With cover
U-bolt		2 sets	U-bolt: 2 M8 nut: 4 Spring washer:4 Flat washer: 4
Instruction manual		1 copy	This document

Description of each part

● Front view



Cover
This cover is used for outdoor installation.

Display part
The measured value and other information are displayed here.

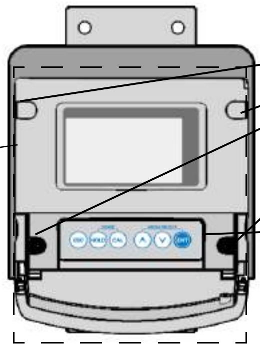
Screw cap
When the two screw caps are removed, the front case screws appear.

Front cover
When the cover is opened, 6 operation keys appear.

Conduit
Up to 6 electric wires of 9 mm to 11.5 mm diameter can be used.

Purge air inlet
Purge air is sent through this inlet to prevent internal corrosion. Normally no connection is required.

With the front cover opened

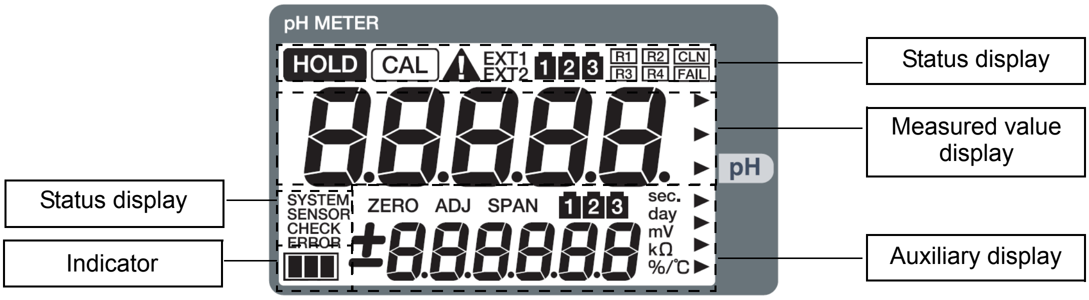


Front case

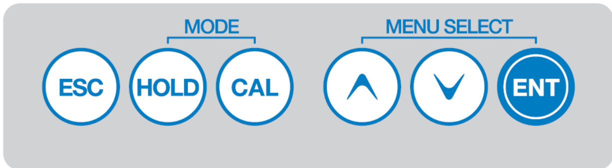
Front case screw
When the four screws are removed, the front case can be opened.

Operation key section
6 operation keys are located.

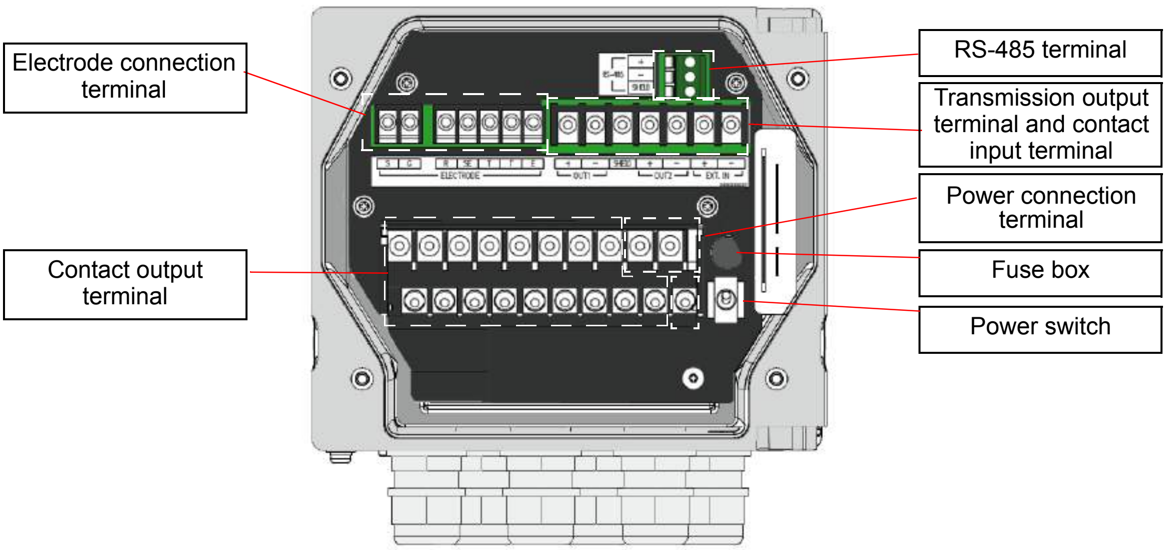
Display part



Operation key section



Terminal block



Description of operation keys

The operation keys are used to change the displayed information, enter parameters, or perform calibration.
The displayed value or item can be changed when it blinks.
Select a blinking value or item with the ▲ and ▼ keys and then press the ENT key. The setting will flash, indicating that it has been established.

<table>
<tr><th>Key</th><th>Nomenclature (meaning) in text</th><th colspan="2">Description</th></tr>
<tr><td rowspan="2">ESC</td><td rowspan="2">ESC key (escape)</td><td colspan="2">Used to go back to the previous menu from any other menu.
During various settings in the Hold mode, pressing this key will go back to the previous item.
The change that has been previously made to any setting is canceled.</td></tr>
<tr><td>Hold down</td><td>Holding down this key in the Hold mode will select the Meas mode.</td></tr>
<tr><td>HOLD</td><td>HOLD key (hold)</td><td>Hold down</td><td>Holding down this key in the Meas mode will select the Hold mode.
Holding down this key in the Hold mode will select the Meas mode.</td></tr>
<tr><td rowspan="2">CAL</td><td rowspan="2">CAL key (calibration)</td><td colspan="2">Allows you to change the number of calibration points or perform calibration again during the basic calibration in the pH cal mode.</td></tr>
<tr><td>Hold down</td><td>Holding down this key until the CAL indicator turns ON will select the pH cal mode.</td></tr>
<tr><td>∧ ∨</td><td>[▲] key
[▼] key
(select)</td><td colspan="2">Used to change the displayed item or value.
Pressing the key will increase/decrease the value by an increment/ decrement of 1. Holding down the key will continuously increase/decrease the value.
The ▲ and ▼ key are used to scroll up and down, respectively. If you excessively scroll up or down with either key, press the other key to return.</td></tr>
<tr><td>ENT</td><td>ENT key (input)</td><td colspan="2">Used to establish each setting or calibrated value.

Note
No change is reflected if you have returned to the previous menu by pressing the ESC key.</td></tr>
</table>

Note

Do not press any operation key with the tip of your nail or any pointed thing.

Description of indicator

Status display

HOLD	HOLD indicator	● Turns ON when the hold mode becomes active. ● Starts blinking when contact input is received or when the hold mode becomes active due to an error
CAL	CAL indicator	Turns ON when the cal mode becomes active.
EXT1	EXT1 indicator	Turns ON when contact input such as the cleaning signal is received
CLN	CLN indicator	Turns ON when the cleaning output contact signal is output
FAIL	FAIL Indicator	Turns ON when the FAIL contact signal is output
SYSTEM ERROR	SYSTEM ERROR indicator	Turn ON when system error E-90, E-91, or E-92 occurs
SENSOR ERROR	SENSOR ERROR indicator	Turns ON when one of sensor errors E-11 to E-72 occurs
SENSOR CHECK	SENSOR CHECK	Illuminated during the check of the temperature sensor type when the temperature sensor type was set to Auto
1 2 3	Bottle indicator	Illuminated during pH calibration ● Upper bottle indicator: The number of calibration points is indicated by bottle indicator number. ● Middle bottle indicator: The current calibration point is indicated by bottle indicator number.

Description of operation modes and menu items

The HP-200 offers 3 kinds of operation modes and 4 kinds of menus.

● Operation mode

● <Meas mode>

In this mode, measurements and control are performed.

● <Cal mode>

In this mode, calibration is performed.
The cal mode is divided into the pH cal mode, the temp cal mode, and the pH manual input cal mode.

● <Hold mode>

In this mode, the measurement and output are stopped to make various settings.

● Menu

The meas and hold modes have the respective menus as shown below:
For selecting each mode or item, see the corresponding paragraph.

● <Meas mode>

<Cal/controlled value menu>

This menu allows you to check the calibrated value or to set/change some controlled values while checking the output.

● <Hold mode>

<Setup menu > (SET)

Prior to operation, this menu allows you to set all the measurement-related parameters such as the input of detector information and the allocation of output.

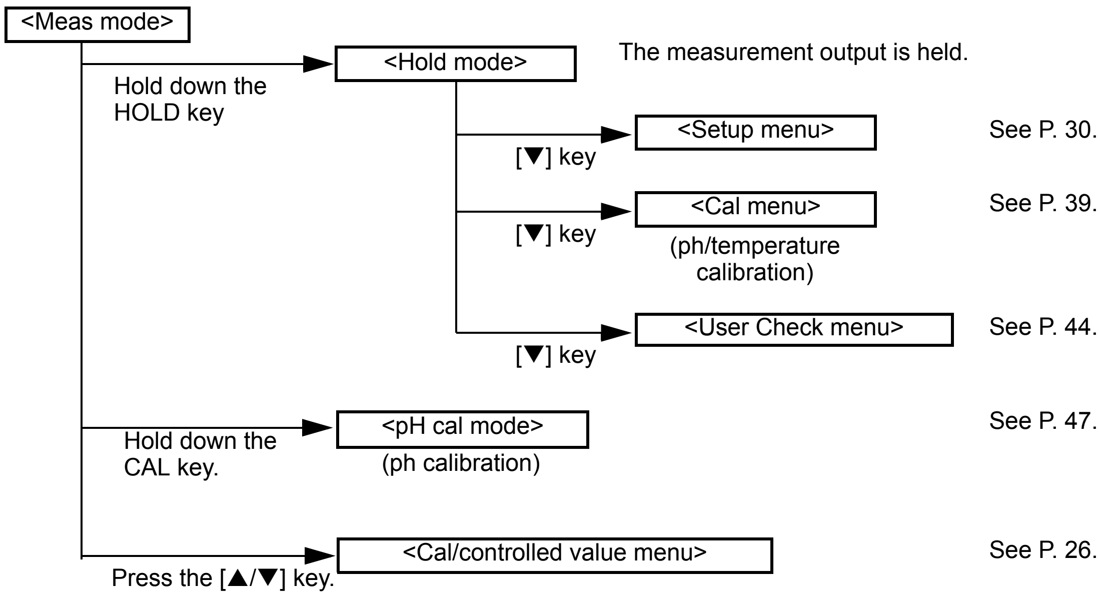
<Cal menu> (CAL)

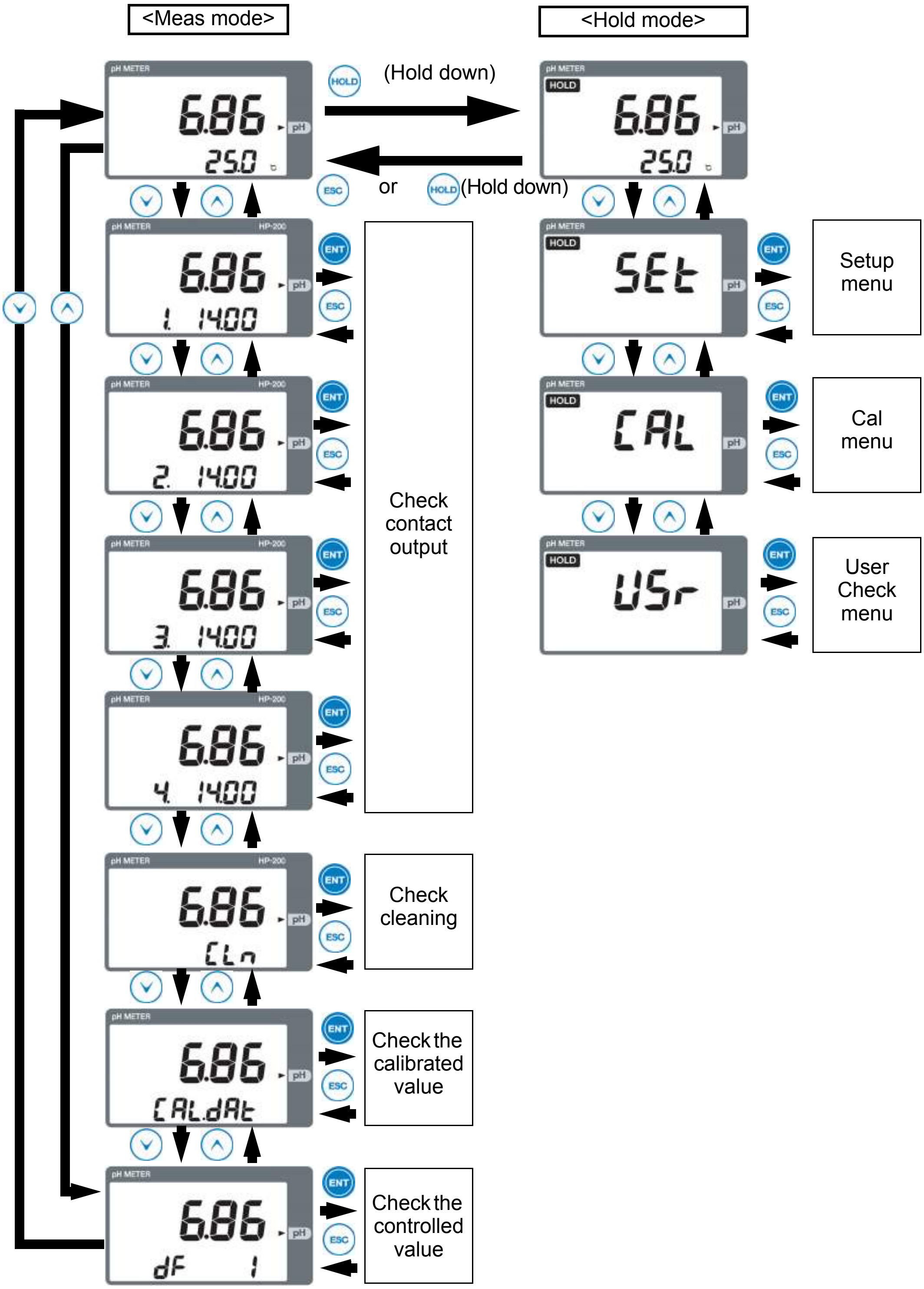
This menu allows you to select the cal mode.

<User Check menu> (USr)

This menu allows you to check the output status, the measured value, and others. The set values can be reset to the factory settings.

Modes and menus avaialble from meas mode




<Meas mode>
<Hold mode>
(Hold down)
or
(Hold down)
Check contact output
Check cleaning
Check the calibrated value
Check the controlled value
Setup menu
Cal menu
User Check menu

Installation

Installation environment

To ensure the use of the HP-200 in stable condition, install the HP-200 in a location where the following requirements are met:

HP-200

- The location shall be well ventilated.
- The ambient temperature shall be between -20°C and 55°C both inclusive.
- Hot air in the location shall be minimal.
- The HP-200 shall not be exposed to direct sunlight.
- The HP-200 shall not be exposed to direct high radiation heat.
- The ambient relative humidity shall be 90% maximum.
- The HP-200 shall not be exposed to splashed water or chemical agent
- The mechanical vibration in the location shall be minimal.
- Maintenance and wire connections shall be possible in the location.
- The location shall be free from dust and corrosive gas.
- The location shall be least affected by an electromagnetic field.
- The altitude shall be 2,000 meters max.
- The variation range of power supply voltage shall fall within ±10%.

Electrode

- The location shall enable the check and maintenance of the electrode.
- The solution under measurement shall be free from air bubbles.
- The solution under measurement shall not corrode the wetted part of the electrode.

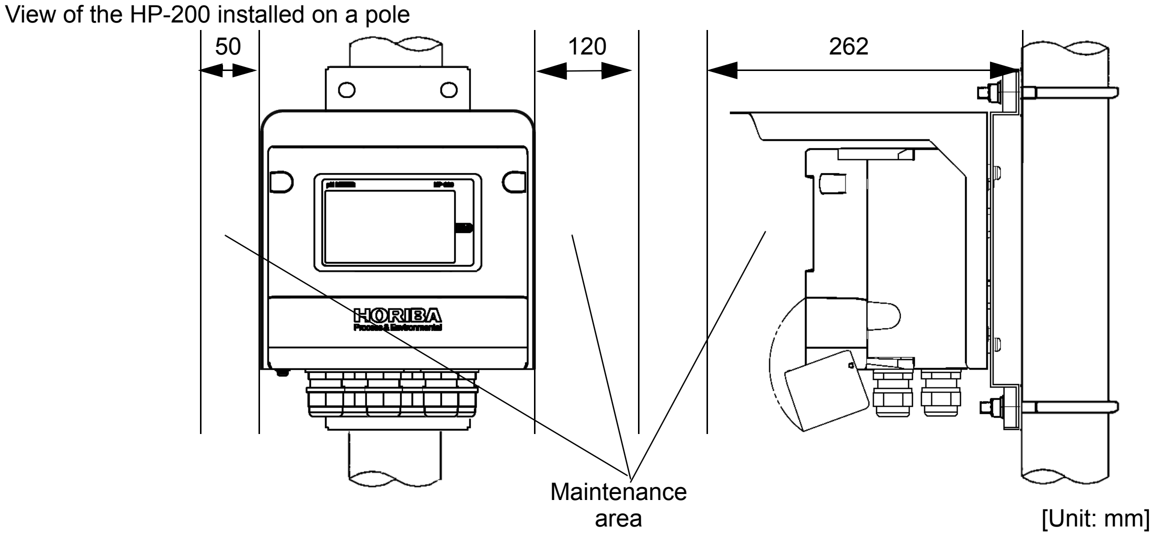
Note

- **For more information, refer to the instruction manual that comes with the electrode.**
- **To measure any solution that has high SS content, the installation of a cleaner is recommended.**

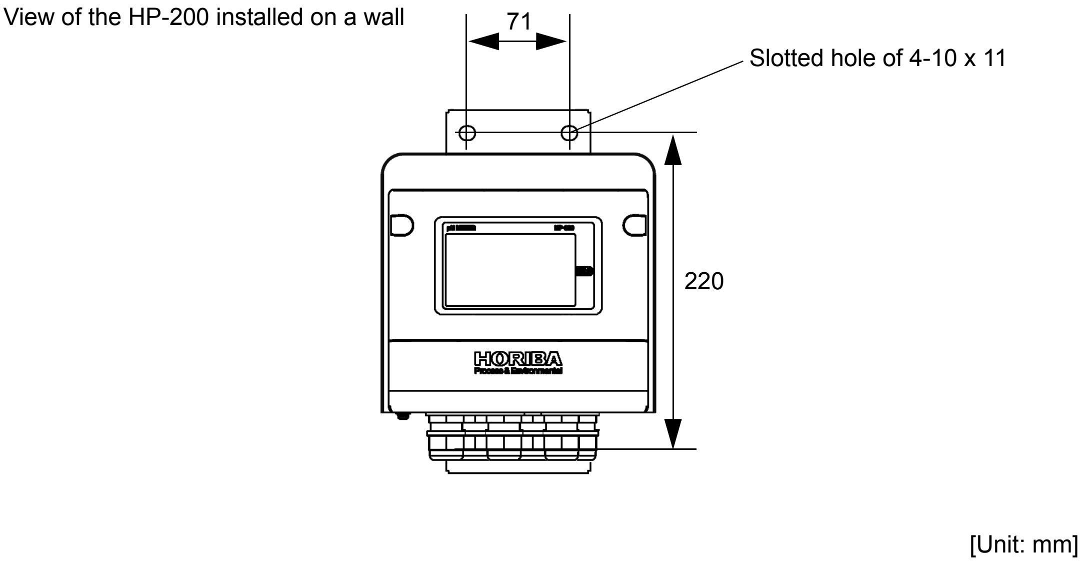
Installing the HP-200

The HP-200 can be installed on a pole (50A) or a wall.

Installation on a pole



Installation on a wall



CAUTION

If there is any corrosive gas in the installation environment, send clean instrumentation air through the air purge inlet on the HP-200.

Note

Provide the necessary area for the maintenance of the HP-200.

Installing the electrode

The electrode can be installed by either of the following two methods:

- Set up the electrode with a holder and then directly immerse the holder in the solution under measurement (immersion type holder).
- Set up the electrode with a holder and then directly insert the holder into the piping line (distribution type holder).

For detailed handling of each holder type, refer to the instruction manual that comes with the product.

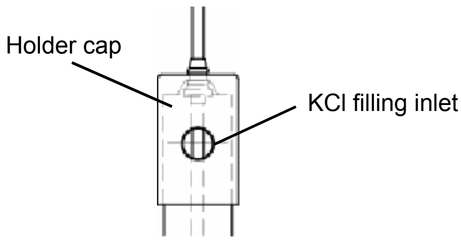
For immersion type holder

1. **Ensure that the electrode tip is always immersed in the solution under measurement even if the solution level fluctuates.**
2. **Ensure that the level of the internal solution (KCl) in the holder is always higher than the level of the solution under measurement.**

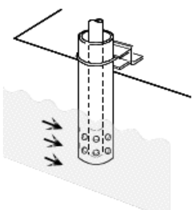
Note

If the level of the internal solution (KCl) in the holder decreases, be sure to add the internal solution.

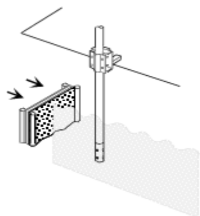
3. **Slightly turn the cap at the top of the holder. The filling inlet will be slightly opened so that large pressure is applied to the inside of the holder.**



4. **The maximum flow rate is 2 m/s. Even if the flow rate is within this limit, the folder may be deformed. Therefore, take the following measures to prevent a decrease in the flow rate.**



Install a pipe of 50 to 100 A that has a introducing hole at its end and then insert the holder into that pipe.



When a partition plate can be installed, e.g., on both sides of a street gutter, install those plates upstream for the benefit of dust removal.

For distribution type holder

1. **Be sure to provide a bypass line on the piping line.**

Note

Unless there is a bypass line, the HP-200 must be stopped when the maintenance or replacement of the electrode is carried out.

2. **Use the distribution type holder with its output side open to atmospheric air.**
3. **Before checking the electrode, close the valve located upstream from the holder. The solution in the holder may flow back.**

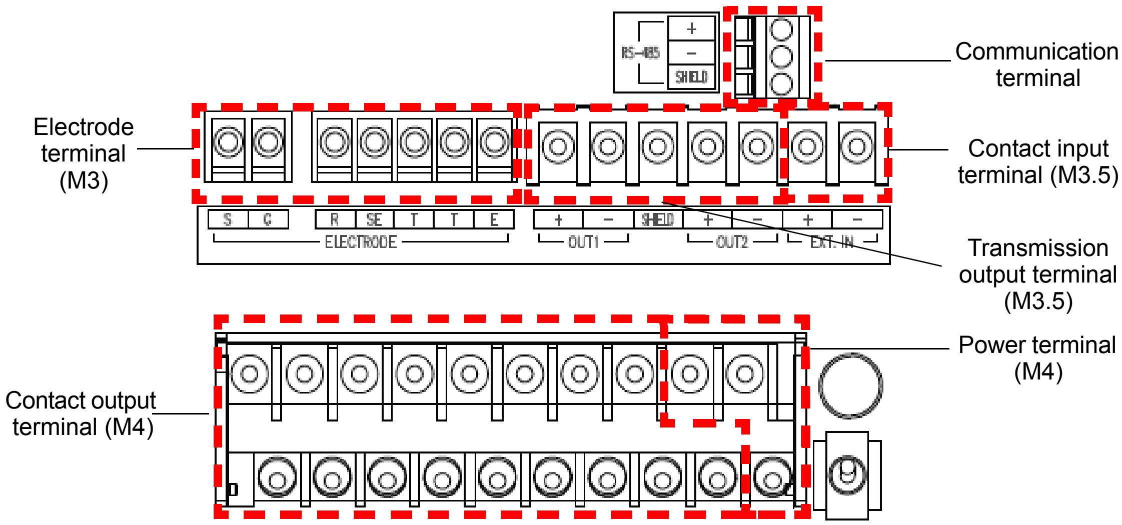
4. An excessively high flow rate may cause fluctuations in the reading. An excessively low flow rate results in response delay. Control the flow rate in accordance with the measurement conditions.
5. For any solution under measurement that has large SS content, install a strainer on the incoming side.

Procedures for connections

● Opening the cover for the HP-200

1. Remove two screw caps located on the upper part of the front of the HP-200. Carefully store the removed caps.
2. Open the front cover.
3. Loosen four screws located on the front of the HP-200. Check that the screws are floated by springs.
4. Open the front case for the HP-200.

● Diagram of connection terminals



Note

- The screws on the front case and the terminal block are structurally prevented from coming off. To connect a terminal, turn the screw counterclockwise until the spring is floated.
- To open the case immediate after it has rained, previously wipe off droplets on the case.

Connecting the power source

- The HP-200 has a power switch.
- For the HP-200, use a free power source for rated voltage of 100 to 240 VAC.
- The terminals crew for the contact output is of M4.
- The applicable electric wire is of 0.75 to 5.5 mm^2 (AWG18 to 10).
- Provide the power switch in a place near the HP-200 so that the power can be turned ON/OFF.

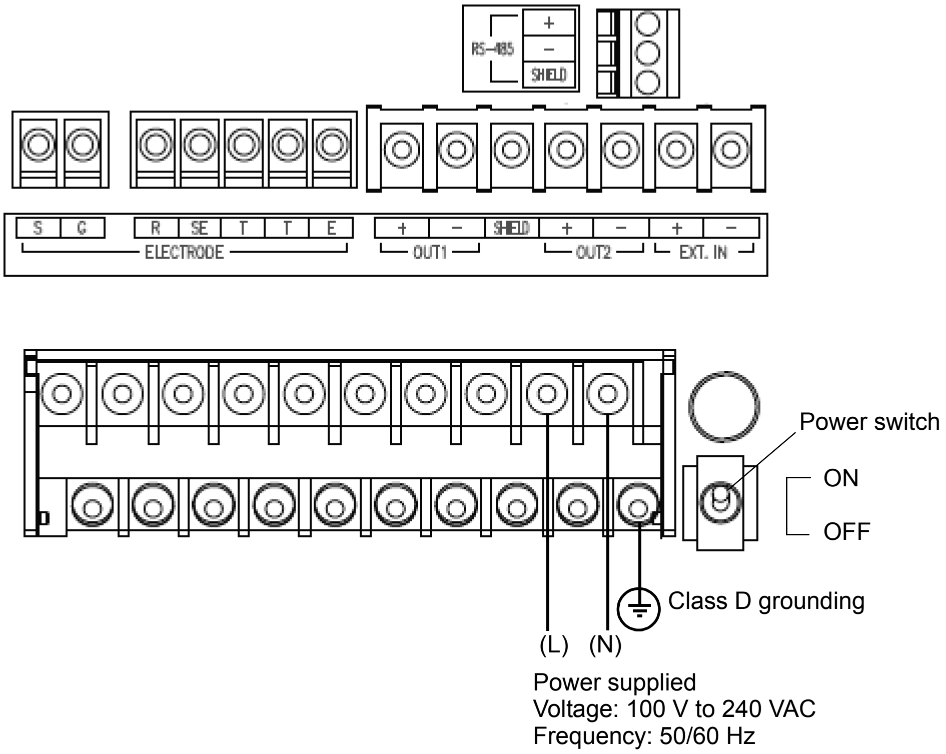
⚠ WARNING

- If lightning might strike, install an arrester on the output side of the HP-200 and on the side of receiving instruments.
- **Electric shock hazard**
  Be sure to check that no power is supplied before starting work.

⚠ CAUTION

- Operation outside the rated range can cause a fault. Therefore, check the power supply voltage. Also check that fluctuations of the power supply voltage fall within ±10%.

As illustrated below, connect the power cable and the grounding cable.



Instructions for grounding

⚠ WARNING

- Be sure to ground the grounding terminal (class D grounding).

Separate this grounding from any other grounding for electric equipment such as a motor.

Connecting the contact output terminal

- The contact capacity is 250 VAC, 3 A maximum or 30 VDC, 3 A maximum for resistance load.
- The terminal screw for the contact output is of M4.
- The applicable electric wire is of 0.75 mm^2 to 5.5 mm^2 (AWG18 to 10).
- If noise is detected from the load, use a varistor or a noise killer.
- Only the CLN output involves voltage from the connected power source. Others are no-voltage contact output.
- For the FAIL output only, NO and NC are reversed.
  When the HP-200 is normal (not in failure), the CF-NOF contact is open and the CF-NCF contact is short-circuited. When the power is OFF, the C-NOF contact is short-circuited.

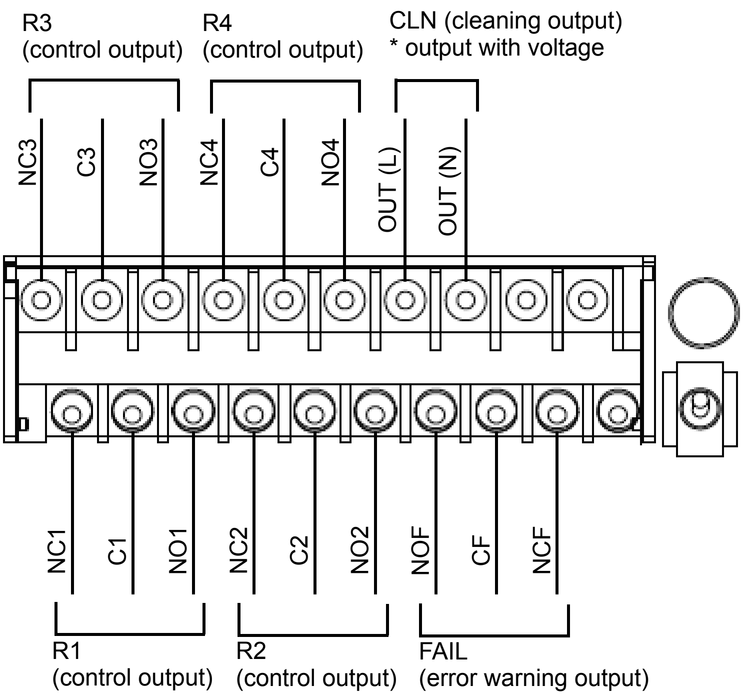
WARNING

To connect any load exceeding the contact capacity or any induction load (e.g., a motor or a pump), be sure to use a power relay exceeding the load rating.

CAUTION

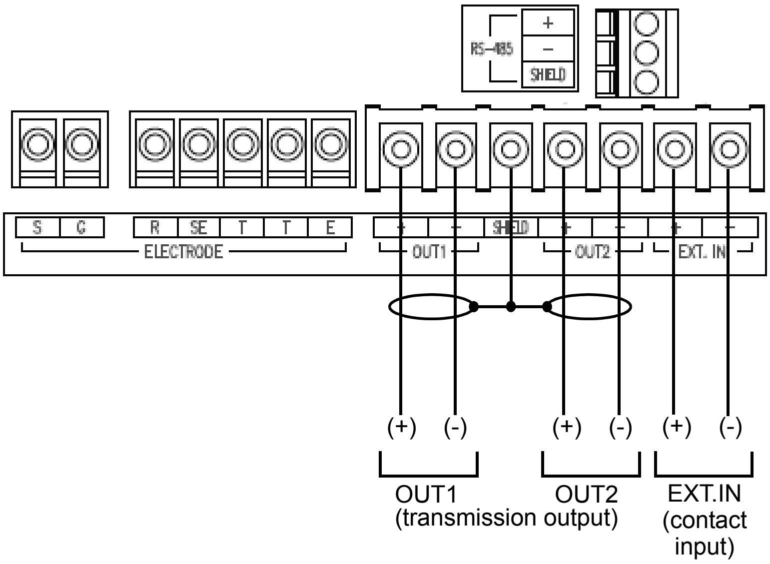
When the HP-200 is OFF, the C-NC contact for R1 to R4 is short-circuited. Therefore, be careful about the connection of load.

As illustrated below, connect the contact output:



Connecting the contact input terminal and the transmission output terminal

- The terminal screws for the contact input and the transmission output are of M3.5.
- The applicable electric wire is of 2 mm^2 (AWG14) maximum.
- For the transmission output cable, use a shielded cable.
- When lightning might strike, install an arrestor on the output side of the HP-200 and on the side of receiving instruments.



Note

- **The resistor for the contact input shall be 100Ω maximum.**
- **The maximum load resistance for the transmission output is 900Ω.**
- **The negative terminal (OUT1) (–) and OUT2 (–) for the transmission output are internally connected and have the same electric potential.**

Tip

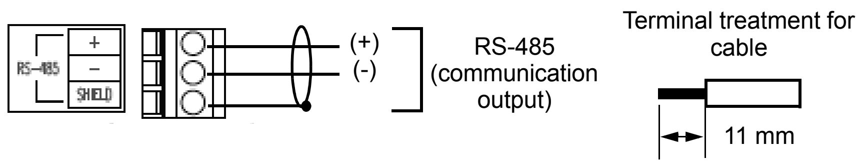
When the contact input is turned ON, the EXT1 indicator on the display part is illuminated.

Connecting the communication terminal

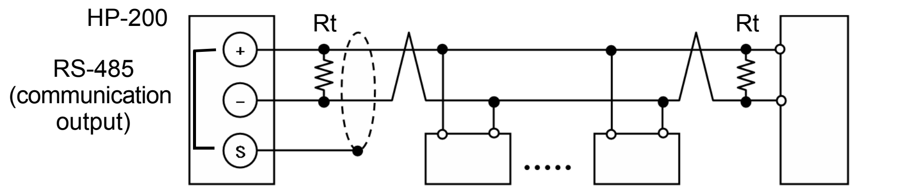
Connecting the RS-485 terminal

The HP-200 has an RS-485 communication terminal. To use this terminal, connect wiring.

- The applicable electric wire is of 0.14 mm^2 to 2.5 mm^2 (AWG26 to 14).
- For the communication output cable, use a twisted shielded pair.
- Up to 32 connections can be made including one for the host computer. For address settings, see “ Set RS-485 address ” (page 38).
- The communication cable length is 500 m maximum.
- To terminate the cable, strip the covering at the end by 11 mm and then connect the stripped end to the terminal.
- Use a terminating resistor (Rt: 120 Ω) for any device at which the RS-485 communication line is terminated.



Example of external connection for communication



RS-485 communication conditions

The RS-485 communication conditions are shown below:

Item	Condition
Baud rate	19200 bps
Character length	8 bit
Parity	None
Stop bit	1 bit

Connecting the electrode cable

Cautionary instructions for electrode cable

The pH electrode cable is highly insulated. In handling this cable, pay attention to the following points:

- Do not wet any cable terminal or the terminal block with water or the like; also do not soil it with dirt, oil, or the like. The insulation will otherwise deteriorate.
  The decreased insulation can cause instable readings. Maintain the electrode cable in a dry, clean state.
  If the electrode cable should be soiled, wipe it off with alcohol or the like and then well dry it.
- In routing the electrode cable, provide sufficient length for the calibration of standard solution and the check and replacement of the electrode.
- Route the electrode cable and the relay cable by avoiding any place near inducing equipment such as a motor and keeping them away from the power cable for such equipment.

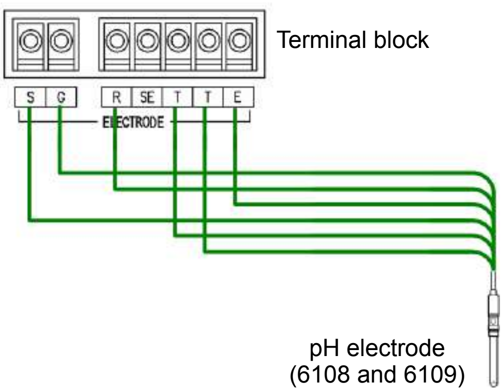
Connections

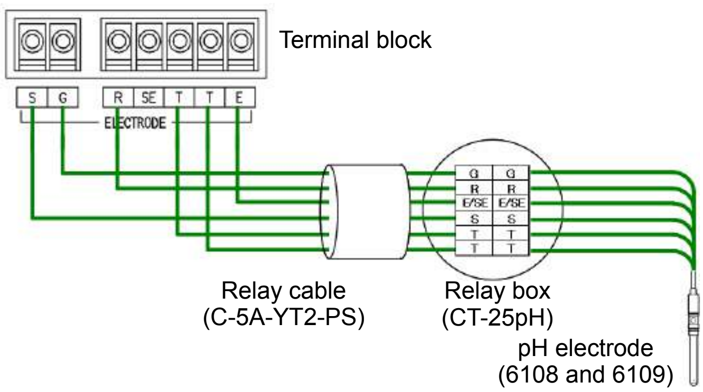
The electrode cable has the following terminals:

S:	Glass electrode shielded drive terminal
G:	Glass electrode terminal
R:	Comparison electrode terminal
SE:	Wetted pole terminal
T, T:	Temperature compensation electrode terminal
E:	Shielded terminal

As illustrated below, connect the terminals of the electrode cable to the corresponding terminals on the terminal block.

- For pH electrodes with S terminal and without SE terminal, such as 6108 and 6109

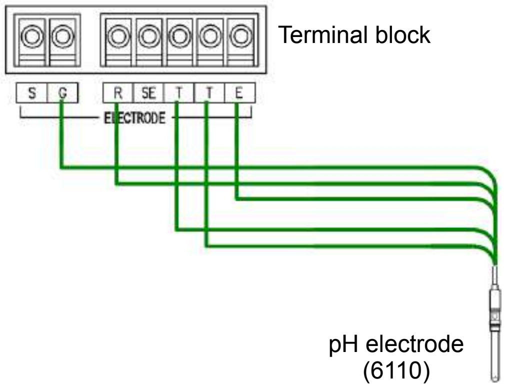


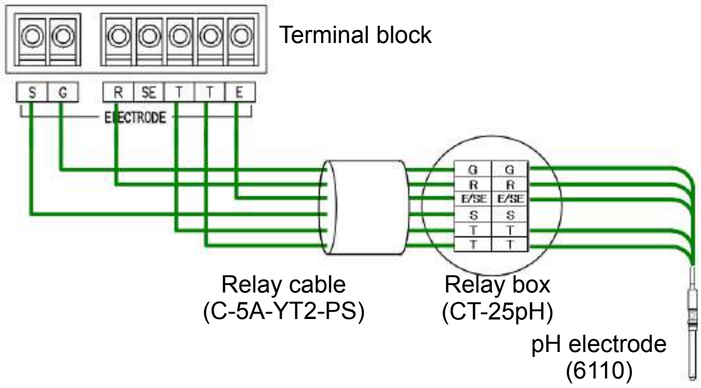


Note

For the above electrode
- **Be sure to select "cLoSE" in “ Set R-SE connection ” (page 32) on the Setup menu.**
- **Be sure to select "g" or "non" in “ Set sensor self-check ” (page 32) on the Setup menu. For this electrode having no wetted pole, a liquid junction resistance error cannot be detected.**

- For pH electrodes without S and SE terminals, such as 6110



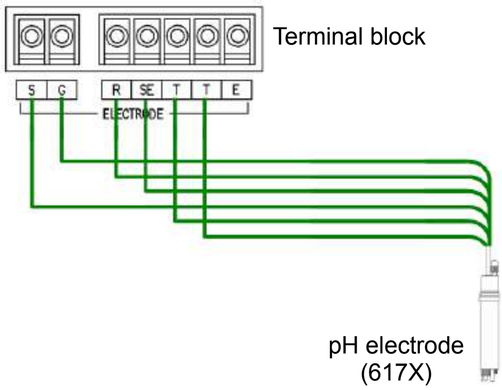


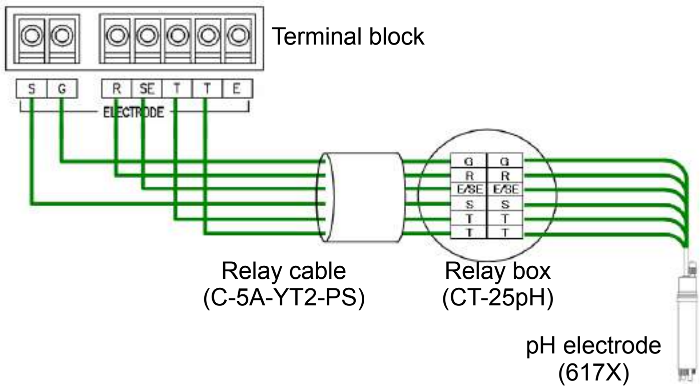
Note

For the above electrode

- **Be sure to select "cLoSE" in “ Set R-SE connection ” (page 32) on the Setup menu.**
- **Be sure to select "g" or "non" in “ Set sensor self-check ” (page 32) on the Setup menu. For this electrode having no wetted pole, a liquid junction resistance error cannot be detected.**

- For pH electrodes with S and SE terminals, such as 6171, 6172, 6173, and 6174





Note

For the above electrode

- **Select "oPEn" in “ Set R-SE connection ” (page 32) on the Setup menu.**
- **A liquid junction resistance error can be detected by selecting "gr" in“ Set sensor self-check ” (page 32) on the Setup menu.**

Temperature compensation electrode

For the HP-200. one of the following four kinds of temperature compensation electrodes can be used:

Resistance value at 25°C: 500 Ω, 680 kΩ, and 10 kΩ

Resistance value at 0°C: 1 kΩ

Check the resistance-temperature detector type for the electrode used and accordingly set the temperature compensation for the HP-200 to an appropriate value.

Reference

“ Setup menu ” (page 30)

Extending the electrode cable

To extend the electrode cable, be sure to use:

- our extension cable (C-5A-YT2-PS or C-5A-YT2-PSE) exclusively for the electrode cable; and
- our relay box (CT-25pH) for exclusive use.

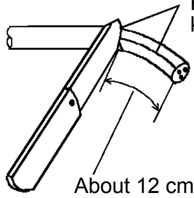
The electrode cable is extendable up to 50 meters between the HP-200 and the electrode.

It is recommended that the dedicated relay cable be housed in a conduit pipe to prevent static electricity from being generated due to induction, vibration, or any other reason. In this case, pass the wiring near the HP-200 through a flexible tube.

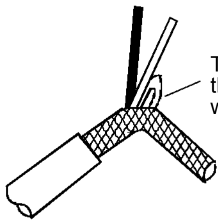
Procedure for terminal treatment of C-5A cable

Perform the following steps to complete the terminal treatment:

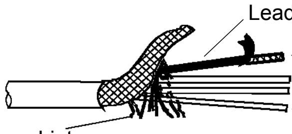
1. **Strip covering of the cable.**



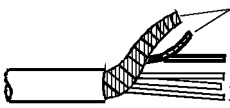
2. **Break the braided shield somewhere in a place near the remaining covering and then take out the lead wires.**



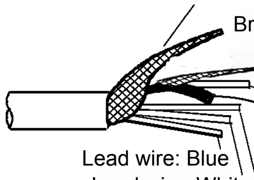
3. **Completely remove lint from the inside of the shield.**



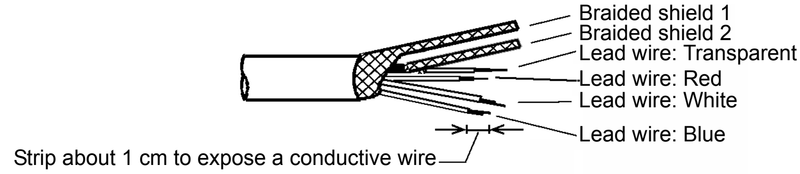
4. **Strip covering of the lead wire (black) up to a place near the remaining covering of the electrode cable and then take out the braided shield for that lead wire.**
5. **Take out the black lead wire with a white line from the braided shield for the lead wire (black).**



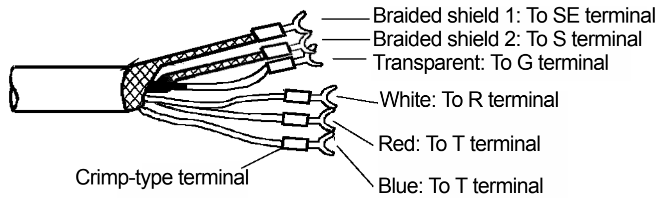
6. **Strip covering of the lead wire (black with a white line).**
   Be sure to strip covering (conductive plastic: black with a white line) up to the root of the transparent lead wire.



7. **Strip each of the lead wires so that its copper wire end is exposed about 1 cm.**
   In stripping the covering of a conductive wire, take care not to cut the conductive wire.
   Cover each of braided shields 1 and 2 with a shrinking tube to avoid short-circuit.



8. **Attach a crimp-type terminal to the conductive wire and then properly crimp it with a crimp tool.**



9. **Pull the terminal to check that it is properly crimped.**

Operation

Preparation for operation

When first using the HP-200 after it has been shipped from factory or reset to the factory settings, perform the following work:

- Checking the wiring
- Initial setup
- Calibration

● Checking the wiring

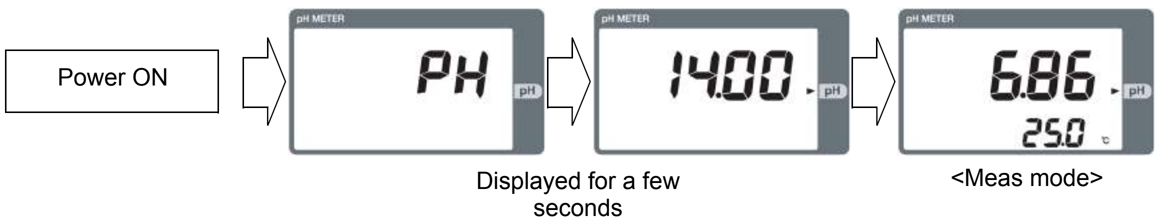
Check the following points:

- The power cable (transmission cable) and the electrode cable are connected properly.
- The terminal block screws are free from loosening.
- The voltage fluctuations fall within the range from 100 V to 240 VAC ±10%.

Starting operation

● Turning ON the power

When the power is turned ON, the initial screen will be displayed with the meas mode selected.



Initial setup

The HP-200 was shipped in the initial state from factory.
Even if the HP-200 is in the initial state, the basic use is possible.
To make any change, follow the instructions given in“ Setup menu ” (page 30)
The initial settings are shown below:

Setup menu item	Screen display	Description	Initial value	Ref. page
Parameter settings	"dF"	Set damping factor.	1	31
	“A.rEt”	Set auto return.	"yES"	31
	“rEt.t”	Set auto return time.	2	31
Sensor settings	“SEnS”	Set sensor type.	"Auto"	32
	“SoL.t”	Set solution temp when temperature sensor is "non."	25	32
	“SELF.C”	Set sensor self-check.	"g"	32
	“r_S.E“	Set R-SE connection	"cLoSE"	32
	“t.CoEF“	Set temp coefficient.	0.000	32
	“t.rEF“	Set temp reference.	25	32
CAL settings	“CAL.t“	Set calibration type.	"Auto"	33
	“CAL.P“	Set calibration point.	2	33
	“A.AtAb“	Set auto stability.	"yES"	33
	“StAb.L“	Set stability level.	"nor"	33
Display settings	“t.diSP“	Set temp display.	"yES"	33
	“S.diSP“	Set sub display.	"t"	33
	“rng.C“	Set range cut.	"yES"	33
Relay output settings	“1.tyPE“	Set contact output type ("r1" or "r2").	"non"	34
		Set contact output type ("r3" or "r4).	"non"	34
	“1.HorL“	Set control method (when the contact output type is "CtrL", "AL", "tPS", "tPF", or "t").	"Hi"	34
	“1.SEt“	Set control value (when the contact output type is "CtrL", "AL", "tPS", "tPF", or "t").	14.00(pH), 100.0(°C)	34
	“1.diFF“	Set control width (when the contact output type is "CtrL").	2.00	34
	“1.dt“	Set contact output delay time (when the contact output type is "AL", "t" or "HoLd").	0	34
	“1.Pb“	Set proportional band (when the contact output type is "tPS" or "tPF").	2.00	34
	“1.Ct“	Set frequency (when the contact output type is "tPS" or "tPF").	10	34
	“1.SFt“	Set control shift amount (when the contact output type is "tPS").	0	34
	“1.Lt“	Set minimum ON time (when the contact output type is "tPS" or "tPF").	2	35
	“1.oL“	Set maximum control amount (when the contact output type is "tPS" or "tPF").	100	35
	“1.F“	Set F zone (when the contact output type is "tPF").	50	35
	“1.ctE“	Set maximum extension time for F zone (when the contact output type is "tPS").	0	35
	“3.CLnt“	Set cleaning output type (when the contact output type is "CLn").	"CLn"	35
Set FAIL output.	“F.SC“	Set FAIL output in a case where the electrode self-check error occurs.	"yES"	35
	“F.oF“	Set FAIL outputin a case where the measured value exceeds the full-scale value.	"no"	35
	“F.tE“	Set FAIL output in a case where the temperature error occurs.	"yES"	35
	“F.dt“	Set FAIL output delay time.	0	35

Setup menu item	Screen display	Description	Initial value	Ref. page
Settings for transmission output 1	“C.rng“	Set current range for transmission output 1	0-14	36
	“rng.0“	Set range zero for transmission output 1.	0.00	36
	“rng.S“	Set range span for transmission output 1.	14.00	36
	“C.HoLd“	Set current hold for transmission output 1.	"HoLd"	36
	“PrES“	Set preset value for transmission output 1.	14.00	36
	“b.out“	Set burnout for transmission output 1.	"non"	36
Settings for transmission output 2.	“rng.0“	Set range zero for transmission output 2.	0.0	37
	“rng.S“	Set range span for transmission output 2.	100.0	37
	“C.HoLd“	Set current hold for transmission output 2.	"HoLd"	37
	“PrES“	Set preset value for transmission output 2.	100.0	37
	“b.out“	Set burnout for transmission output 2.	"non"	37
Setting for cleaning output.	“CLn“	Set cleaning output.	"yES"	38
	“tyPE“	Set cleaning output type.	"or"	38
	“CyCLE“	Set cleaning output frequency (When cleaning output is set to "yES" and cleaning output type "nor", "or", or "And").	2.0	38
	“CLn.t“	Set cleaning output time (when cleaning output is set to "yES").	10	38
	“HoLd.t“	Set preset value after finishing cleaning.	120	38
RS485 settings	“Addr“	Set address for RS-485.	0	38

To exit the setup menu, hold down the ESC key until the HOLD indicator turns OFF. This will select the meas mode.

Calibration

See the following pages to perform the pH calibration and the temperature calibration.

- “ pH calibration ” (page 47)
- “ Temperature calibration ” (page 53)

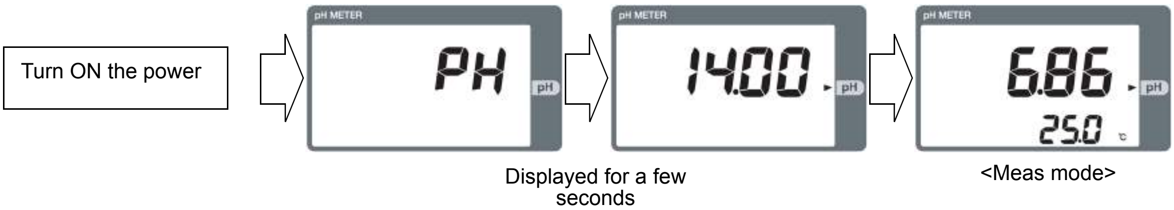
Measurement

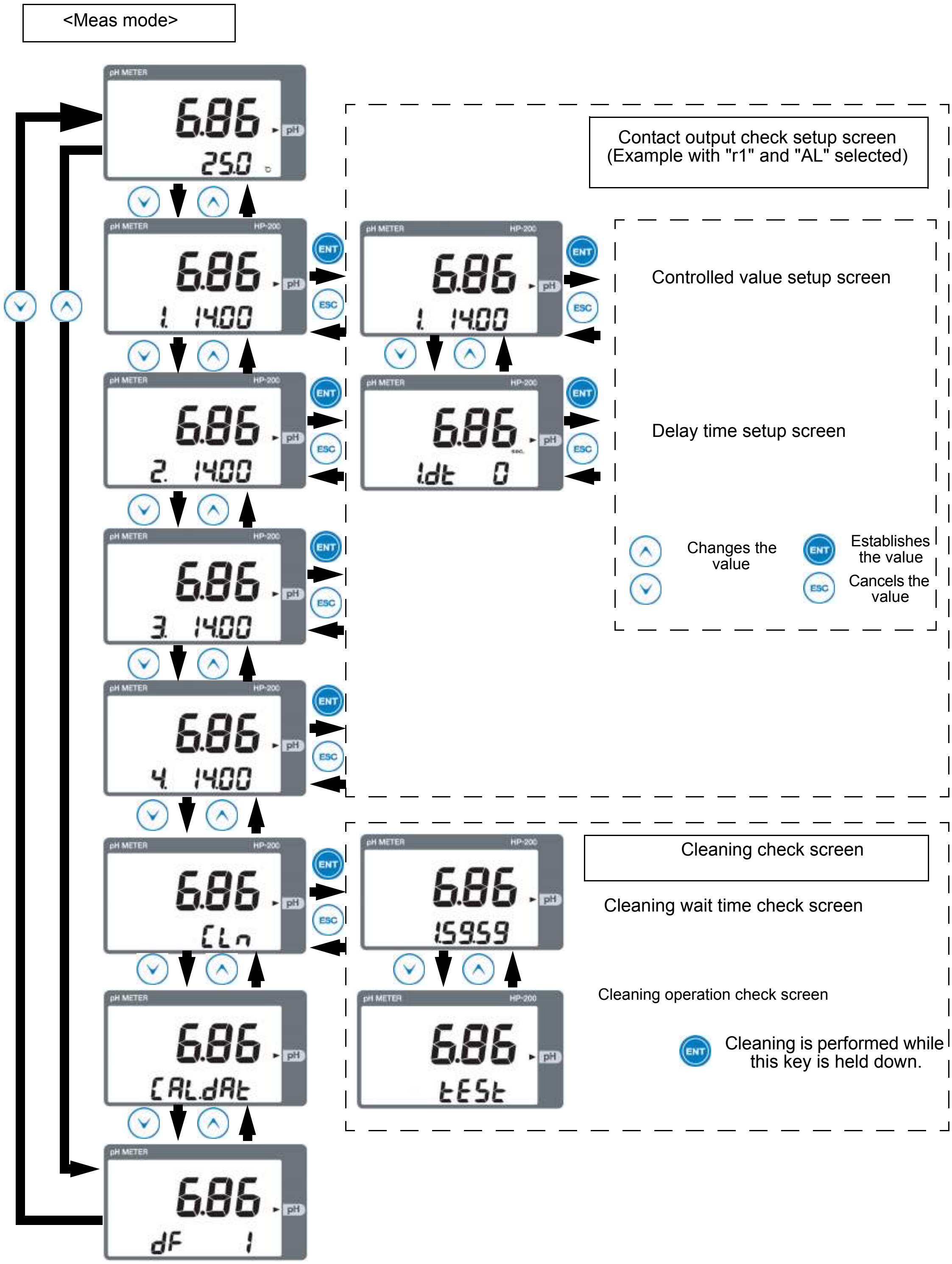
Meas mode

1. **Turn ON the power.**
   The measurement object is displayed on the display part.
   A measurement range is also displayed with the Meas mode selected.
   The measurement is started with the measured value displayed.
   In the Meas mode, this is the normal state.
2. **The screen is changed by pressing the [▲/▼] key.**
   Changing the screen allows you to change the moving average number or check calibration data.

Note

Prior to measurement, be sure to calibrate the electrode.
See “ Calibration ” (page 47)


Turn ON the power
pH METER
PH
pH
1400
6.86
25.0
Displayed for a few seconds
<Meas mode>


<Meas mode>
pH METER
HP-200
Contact output check setup screen (Example with "r1" and "AL" selected)
Controlled value setup screen
Delay time setup screen
Changes the value
Establishes the value
Cancels the value
ENT
ESC
1. 1400
2. 1400
3. 1400
4. 1400
1dt 0
CLn
CALdAt
dF 1
Cleaning check screen
Cleaning wait time check screen
15959
Cleaning operation check screen
tESt
Cleaning is performed while this key is held down.

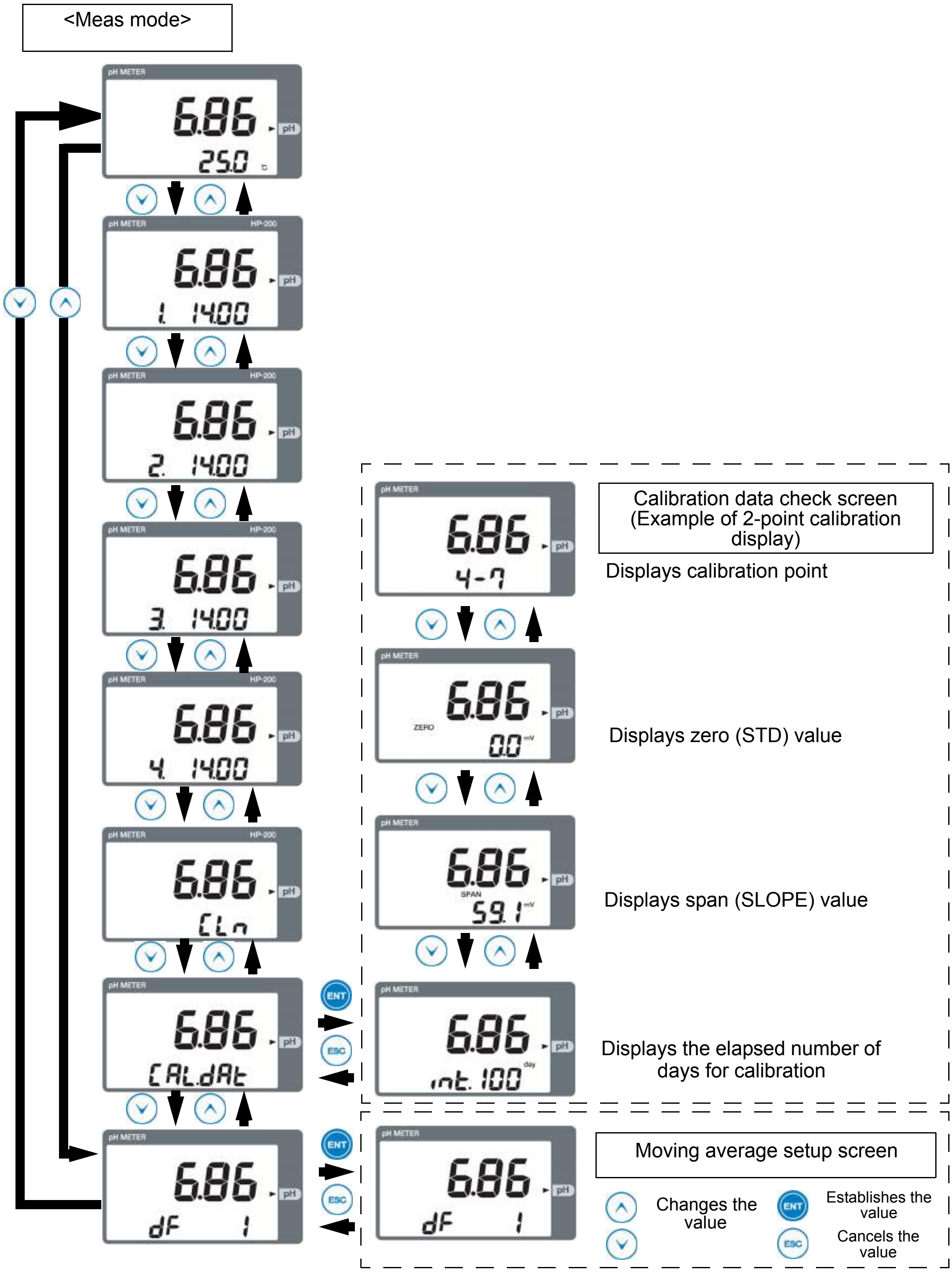

<Meas mode>
Calibration data check screen
(Example of 2-point calibration display)
Displays calibration point
Displays zero (STD) value
Displays span (SLOPE) value
Displays the elapsed number of days for calibration
Moving average setup screen
Changes the value
Establishes the value
Cancels the value

Table 1 Items displayed in Meas mode

Screen 1	Screen 2	Condition	Description	Action	Remarks
		With temperature display	Measured value display screen	The measured values for pH and temperature are displayed.	The temperature may not be displayed depending on the setting.
		When rly type is set to "non"	Relay condition setup screen (the number at the lower left part indicates relay number)		
		When "CtrL", "AL", "tPS", "tPF", or "t" is selected	Relay condition setup screen (the number at the lower left part indicates relay number)	The set value (pH or temperature) is displayed. The set value can be changed.	Input range ● pH: 0.00 to 14.00 ● temp: –10.0 to 120.0 (°C)
		When "CtrL" is selected	Control range value screen	The control range is displayed or set using pH values	Input range ● pH: 0.02 to 4.00
		When "AL" or "t" is selected	Delay time screen	The delay time is displayed or set in seconds.	Input range ● 0 to 600 seconds
		When "tPS" or "tPF" is selected	Proportional band value screen	The proportional band value for time-sharing proportional control is displayed or set using a pH value.	Input range ● pH: 0.02 to 4.00
		When "tPS" or "tPF" is selected	Frequency time screen	The frequency time is displayed or set in seconds.	Input range ● 5 to 300 seconds
		When "tPS" is selected	Control shift value screen	The shift amount for control output in time-sharing proportional control is displayed or set in percentage.	Input range ● 0 to 50(%)
		When "tPF" is selected	F-zone value setting	The frequency (F-zone) in time-sharing proportional control is displayed or set in percentage.	Input range 1 to 100(%)
		When "tPF" is selected	Frequency extension time screen	The maximum for additional OFF time extended by the F-zone function is displayed or set in seconds.	Input range ● 0 to 300 seconds

Screen 1	Screen 2	Condition	Description	Action	Remarks
		When "HoLd" is selected	Delay time screen	The delay time is displayed or set in seconds.	Input range ● 0 to 600 seconds
		When "CLn" is selected	Output during cleaning screen	Indicates that data is output during cleaning and in the hold mode after cleaning.	
		When "CLn.t" is selected	Trigger output during cleaning screen	Indicates that the trigger signal is output after cleaning has been finished.	
			Cleaning output wait time screen	Displays the time up to the start of the next cleaning in the format of "hours.minutes.seconds."	When the wait time is 10 hours or more, it is displayed in the format of "hours.minutes."
			Cleaning output test operation screen	Cleaning is performed while the ENT key is held down.	When the [ENT] key is released, the output will be held during the HOLD time after cleaning.
		Without calibration	No calibration data screen	Indicates that there is no past calibration data.	Initial value ● STD: 0.0 mV ● SLOPE: 59.1 mV
		During one-point calibration	Calibration data check screen Calibration point check screen	Displays the calibration points for calibration data.	
			Zero (STD) value check screen	Indicates asymmetry potential (STD) at pH7 as electric potential (mV).	
			Span (SLOPE) value check screen	Indicates the sensitivity (SLOPE) as electric potential per 1 pH.	
			Elapsed days after calibration screen	Indicates the number of elapsed days after the last calibration.	Note: The elapsed time is accumulated only when the power is ON.

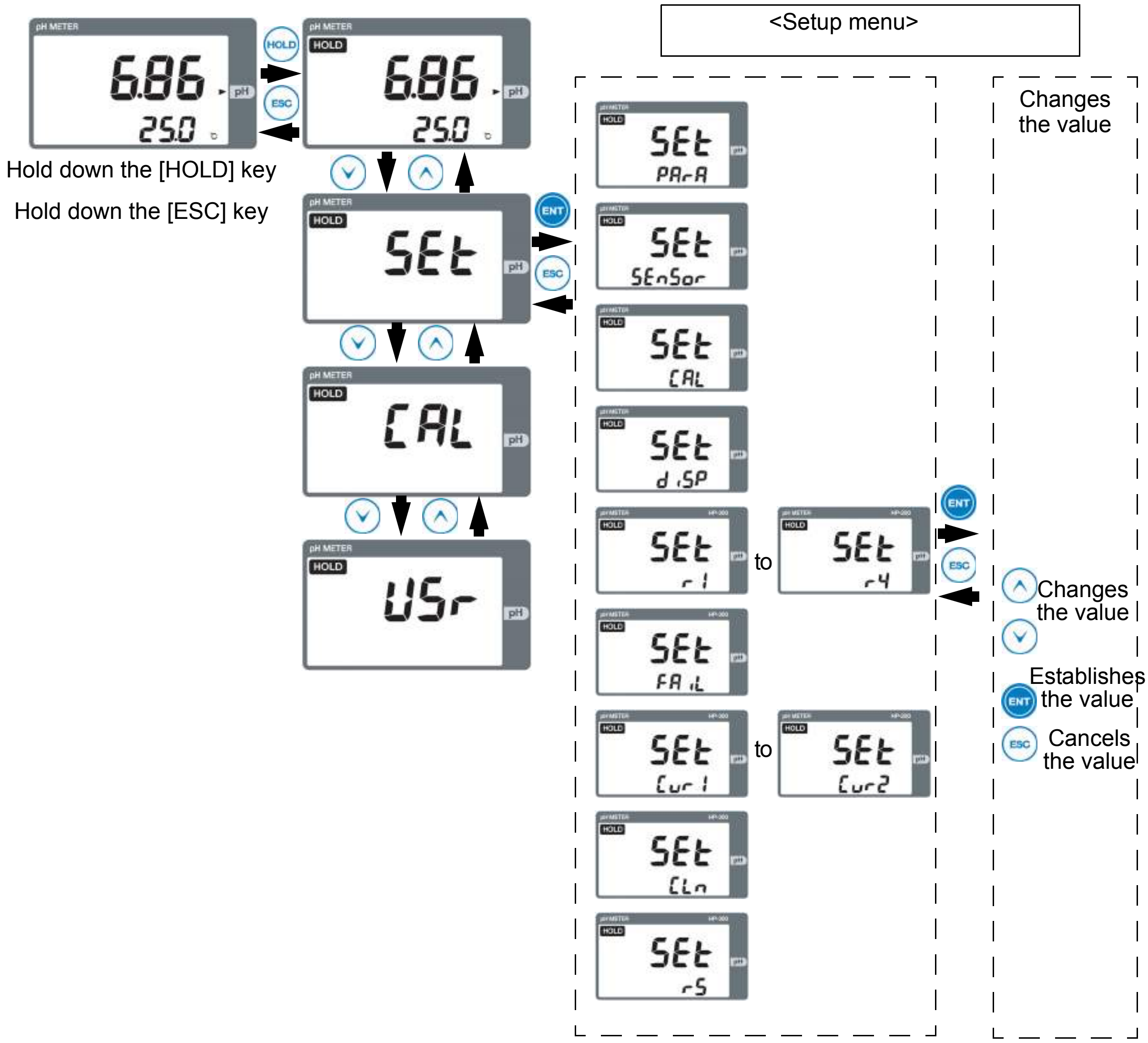
Screen 1	Screen 2	Condition	Description	Action	Remarks
		During two-point calibration	Calibration data check screen Calibration point check screen	Indicates the calibration points for calibration data (this display example assumes two-point calibration)	Note: For manual input ("PH.Adj"), "Adj" is displayed on the display part.
			Zero (STD) value check screen	Indicates asymmetry potential (STD) at pH7 as electric potential (mV).	
			Span (SLOPE) value check screen	Indicates the sensitivity (SLOPE) as electric potential per 1 pH.	
			Elapsed days after calibration screen	Indicates the number of elapsed days after the last calibration.	Note: The elapsed time is accumulated only when the power is ON.
		During three-point calibration	Calibration data check screen Calibration point check screen	Indicates the calibration points for calibration data (this display example assumes three-point calibration).	
			Zero (STD) value check screen	Indicates asymmetry potential (STD) at pH7 as electric potential (mV).	
			Span (SLOPE1) value check screen	Indicates the sensitivity (SLOPE) for span 1 (SLOPE1) as electric potential per 1 pH.	
			Span (SLOPE2) value check screen	Indicates the sensitivity (SLOPE) for span 2 (SLOPE2) as electric potential per 1 pH.	
			Elapsed days after calibration screen	Indicates the number of elapsed days after the last calibration.	Note: The elapsed time is accumulated only when the power is ON.
			Moving average count check screen (damping factor)	Allows you to check or set the moving average count.	Note: ● Change this parameter when the measured value is not stable enough. ● If an error occurs during this setting, you will exit this screen with the previous setting maintained.

Setup menu

The Setup menu allows you to specify your electrode type, measurement conditions, display parameters, calibration method, and transmission output.

● Entering the Setup menu

1. **In the meas mode, hold down the HOLD key until the [HOLD] indicator turns ON.**
2. **Press the [▲/▼] key to display "SEt" on the display part for the measured value.**
3. **[Press the ENT key.**
4. **Press the [▲/▼] key to display a desired item on the auxiliary display part.**
5. **Press the [ENT] key.**



Note

The displayed value changes while the HOLD indicator is iluminated. For the transmission output, however, the output value specified on the Setup menu is displayed.
See " Set currently held value for transmission output 1 " (Page 36) " Set currently held value for transmission output 2 " (Page 37)

Table 2 <Items displayed on Setup menu

Screen 1	Screen 2	Description	Setting range	Initial value	Remarks
		Set the moving average count (damping factor).	1 to 50	1	● Set the moving average count. ● Change the value when the measured value is not stable enough.
		Set auto return	"yES", "no"	"yES"	This setting determines whether the hold mode automatically returns to the meas mode. Selecting "yES" enables auto return. Note: Auto return does not occur during any setting change or calibration even if "yES" is selected.
		Set auto return time.	1 to 24(h)	2	This setting determines the time for auto return from the hold mode to the meas mode. Enter hours. Note: This setting is only displayed when auto return is set to "yES."

Screen 1	Screen 2	Description	Setting range	Initial value	Remarks
		Set sensor type	"non", "Auto", 500, 1k 6.8k, 10k (Ω)	"Auto"	Selecting "Auto" automatically determines the temperature sensor type. Find your temperature sensor type in the instruction manual for the electrode and then specify it accordingly. Note: ● When no temperature electrode is used, set this parameter to "non." ● When the temperature exceeds the display range with "Auto" selected, the sensor type is automatically detected again, resulting in possible display of wrong temperature. In this case, manually specify your sensor type.
		Set solution temperature when "non" is selected for temperature sensor type	0 to 100 (°C)	25	Set the solution temperature when no temperature electrode is used. Note: This setting is only displayed when temperature sensor type is set to "non.
		Set sensor self-check	"non", "g", "gr"	"g"	To enable the sensor self-check, use this parameter. ● "g": Only checks for breakage of response membrane ● "gr": Checks for breakage of response membrane plus liquid junction error ● "non": Performs no self-check Note: "gr" is only available for the electrode with liquid grounding.
		Set R-SE connection	"cLoSE", "oPEn"	"cLoSE"	This setting determines whether electrode terminals R and SE are connected inside the HP-200. ● "cLoSE": Shorts the circuit between R and SE. ● "oPEn": Opens the circuit between R and SE. Note: Set this parameter to "oPEn" for only the electrode with liquid grounding.
		Set temp coefficient	–0.100 to 0.100 (pH/°C)	0.000	Specify a solution temperature coefficient. Specify a pH value that changes per 1°C. ● This parameter is set to 0.0000, normally requiring no change.
		Set temp reference	5 to 95 (°C)	25	Specify a reference temperature for solution temperature compensation. ● This parameter is set to 25°C, normally requiring no change.

Screen 1	Screen 2	Description	Setting range	Initial value	Remarks
		Set calibration type	"Auto", "bASic"	"Auto"	Select a calibration method. ● Normally "Auto" should be selected. For three-point calibration or calibration using a special standard solution, set this parameter to "bASic.
		Set calibration point	1,2,3	2	Specify a calibration point. Note: This setting is only displayed when the calibration type is "bASic."
		Set auto stability	"yES", "no"	"yES"	Specify whether the automatic stability detection is enabled.*1 Note: This setting is only displayed when the calibration type is "bASic.
		Set stability level	"Lo", "nor", "Hi"	"nor"	Select an automatic stability detection level. ● For higher stability detection level, select "Hi"; for lower one, select "Lo." *1) Note: This setting is only displayed when calibration type is set to "bASic" and automatic stability detection to "yES.
		Set temp display	"yES", "no"	"yES"	This setting determines whether temperature is displayed on the auxiliary display part.
		Set sub-display	"non", "r1", "r2", "r3", "r4", "t"	"t"	This setting determines what is displayed on the auxiliary display part when the power is turned ON or when the hold mode is changed over to the meas mode. Note: The solution temperature is displayed when sensor type is set to "non" or "t."
		Set range cut	"yES", "no"	"yES"	This setting determines whether the display is hidden when the measured value for pH or temperature exceeds the display range. Display range ● "yES": pH: 0 to 14 Temperature: 0to100(°C) ● "no": pH: −1 to 15 Temperature: −10 to 110 (°C) (when sensor type is set to "Auto") Temperature: −20 to 130 (°C) (when sensor type is set to "Auto")

Screen 1	Screen 2	Description	Setting range	Initial value	Remarks
   		Set contact output type ("r1" or "r2")	"non", "CtrL", "AL", "tPS", "tPF", "t"	"non"	Specify a contact output type. ● "non": No output ● "CtrL": Control output ● "AL": Warning output ● "tPS": Time-sharing proportional control (shift function) ● "tPF": Time-sharing proportional control (F-zone function) ● "t": Temperature warning output
		Set contact output type ("r3" or "r4")	"non", "CtrL", "AL", "t", "HoLd", "CLn"	"non"	Specify a contact output type. ● "non": No output ● "CtrL": Control output ● "AL": Warning output ● "t": Temperature warning output ● "HoLd": Output during hold mode ● "CLn": Output during cleaning (including the hold time after cleaning)
		Set control method (when contact output type is set to "CtrL", "AL", "tPS", "tPF", or "t")	"Hi", "Lo"	"Hi"	Specify whether control is performed by upper limit operation or lower limit operation. ● "Hi": Performs control by upper limit operation. ● "Lo": Performs control by lower limit operation Note: This setting is only displayed when contact output type is set to "CtrL", "AL", "tPS", "tPF", or "t."
		Set controlled value (when contact output type is set to "CtrL", "AL", "tPS", "tPF, or "t")	0.00 to 14.00 (pH) −10.0 to 120.0 (°C)	14.00 (pH), 100.0 (°C)	Specify a controlled value. Note: This setting is only displayed when contact output type is set to "CtrL", "AL", "tPS", "tPF, or "t.
		Set control width (when contact output type is set to "CtrL")	0.02 to 4.00 (pH)	2.00	Specify control width. Note: This setting is only displayed when contact output type is set to "CtrL."
		Set contact output delay time (when contact output type is set to "AL", "t", or "Hold")	0 to 600 (s)	0	Specify the time in seconds between warning and contact output. Note: This setting is only displayed when contact output type is set to "AL", "t", or "Hold. The delay time up to the cancellation of warning is 0 seconds.
		Set proportional band (when contact output type is set to "tPS" or "tPF")	0.02 to 4.00 (pH)	2.00	Specify proportional bandwidth using a pH value. Note: This setting is only displayed when contact output type is set to "tPS" or "tPF.
		Set frequency (when contact output type is set to "tPS" or "tPF")	5 to 300 (s)	10	Specify frequency time in seconds. Note: This setting is only displayed when contact output type is set to "tPS" or "tPF."
		Set control shift amount (when contact output type is set to "tPS")	0 to 50 (%)	0	Specify a control shift amount in percentage. Note: This setting is only displayed when contact output type is set to "tPS.

Screen 1	Screen 2	Description	Setting range	Initial value	Remarks
   		Set minimum ON time (when contact output type is set to "tPS" or "tPF")	2 to 10 (s)	2	Specify minimum ON time in seconds. Note: This setting is only displayed when contact output type is set to "tPS" or "tPF."
		Set maximum control amount (when contact output type is set to "tPS" or "tPF")	50 to 100 (%)	100	Specify the maximum control amount in percentage. Note: This setting is only displayed when contact output type is set to "tPS" or "tPF."
		Set F-zone amount (when contact output type is set to "tPF")	1 to 100 (%)	50	Specify an F-zone amount in percentage. Note: This setting is only displayed when contact output type is set to "tPF."
		Set maximum extension time for F-zone (when contact output type is set to "tPF")	0 to 300 (s)	0	Specify maximum extension time for F-zone in seconds. Note: This setting is only displayed when contact output type is set to "tPF." When this parameter is set to 0 seconds, the F-zone function is disabled.
		Set cleaning output type (when contact output type is set to "CLn")	"CLn", "trg"	"CLn"	Specify a cleaning output type. ● "CLn": Performs contact output during cleaning (including the hold time after cleaning). ● "trg": Performs contact output 2 and 5 seconds after cleaning. Note: This setting is only displayed when contact output type is set to "CLn."
		Set FAIL output in case of electrode self-check error	"yES", "no"	"yES"	Specify whether the FAIL signal is output when electrode self-check error (E-71 or 72) occurs.
		Set FAIL output in case where measured value exceeds full-scale value	"yES", "no"	"no"	Specify whether the FAIL signal is output when the measured value exceeds the full-scale value.
		Set FAIL output in case of abnormal temperature error	"yES", "no"	"yES"	Specify whether the FAIL signal is output when abnormal temperature error (E-21, 22, or 25) occurs.
		Set FAIL output delay time	0 to 600 (s)	0	Specify the time between FAIL output and contact output in seconds. Note: The delay time for cancellation of FAIL output is 0 seconds.

Screen 1	Screen 2	Description	Setting range	Initial value	Remarks
		Set current range for transmission output 1	"opt" 0-14, 0-10, 0-8, 2-12, 3-11, 4-14, 4-10, 6-14,	0-14	Specify a transmission output range for transmission output 1. Note: An arbitrary value may be specified when transmission output range is set to "opt."
		Set range zero for transmission output 1	−1.00 to 15.00 (pH)	0.00	Specify a pH value for 4 mA when transmission output 1 is set to "opt. Note: This setting is only displayed when transmission output range is set to "opt."
		Set range span for transmission output 1	−1.00 to 15.00 (pH)	14.00	Specify a pH value for 20 mA when transmission output 1 is set to "opt. Note: This setting is only displayed when transmission output range is set to "opt."
		Set currently held value for transmission output 1	"CAL", "HoLd", "PrES"	"HoLd"	Specify a transmission output value for transmission output 1 (in the hold mode). ● "CAL": Directly outputs the calibration value during calibration. ● "HoLd": Outputs the directly previous value while holding it. ● "PrES": Outputs the preset value.
		Set preset value for transmission output 1	−1.00 to 15.00 (pH)	14.00	Specify an arbitrary output value for transmission output in the hold mode. Note: This setting is only displayed when preset value is set to "PrES.
		Set burnout value for transmission output 1	"non", "out.4", "out.20"	"non"	Specify output during burnout for transmission output 1. ● "out.4": Outputs 3.8 mA during burnout. ● "out.20": Outputs 21 mA during burnout. ● "non": Outputs none during burnout. Note: The burnout output only occurs in the Meas mode.

Screen 1	Screen 2	Description	Setting range	Initial value	Remarks
		Set range zero for transmission output 2	–20.0 to 130.0 (°C)	0.0	Specify a temperature value for 4 mA for transmission output 2.
		Set range span for transmission output 2	–20.0 to 130.0 (°C)	100.0	Specify a temperature value for 20 mA for transmission output 2.
		Set currently held value for transmission output 2	"CAL", "HoLd", "PrES"	"HoLd"	Specify a transmission output value for transmission output 2 in the hold mode. ● "CAL": Directly outputs a calibration value during calibration. Otherwise outputs the directly previous value while holding it. ● "HoLd": Outputs the directly previously value while holding it. ● "PrES": Outputs the preset value.
		Set preset value for transmission output 2	–20.0 to 130.0 (°C)	100.0	Specify an arbitrary output value for transmission output 2 in the hold mode. Note: This setting is only displayed when currently held value is set to "PrES.
		Set burnout output for transmission output 2	"non", "out.4", "out.20"	"non"	Specify output during burnout for transmission output 2. ● "out.4": Outputs 3.8 mA during burnout. ● "out.20": Outputs 21 mA during burnout. ● "non": Outputs none during burnout. Note: The burnout output only occurs in the Meas mode.

Screen 1	Screen 2	Description	Setting range	Initial value	Remarks
		Set cleaning output	"yES", "no"	"yES"	Specify whether cleaning output is enabled or disabled. ● "yES": Enables cleaning output. ● "no": Disables cleaning output.
		Set cleaning output type	"nor", "or", "And", "trg"	"or"	Specify a cleaning output type. ● "nor": Provides cleaning output in accordance with the cleaning output frequency and the cleaning time. ● "or": Provides cleaning output in accordance with the cleaning output frequency and the cleaning time. Also provides it when an external contact signal is input. ● "And": Provides cleaning output in accordance with the cleaning output frequency and the cleaning time only when an external contact signal is input. ● "trg": Provides cleaning output in accordance with the cleaning time when an external cleaning signal is input as trigger for 2 seconds minimum. Note: After cleaning output has been finished, the specified held value is output in accordance with the corresponding output time.
		Set cleaning output frequency (when cleaning output is set to "yES" and cleaning output type to "nor", "or", or "And")	0.1 to 168.0(h)	2.0	Specify a cleaning frequency for cleaning output in hours.
		Set cleaning output time (when cleaning output is set to "yES")	2 to 600(s)	10	Specify cleaning time for cleaning output in seconds.
		Set held value output duration after cleaning	2 to 600(s)	120	Specify duration when the output value is held after cleaning has been finished. Note: When cleaning output is set to "no," specify duration when the output value is held after test cleaning has been finished.
		Set RS-485 address	0 to 99	0	Specify an address for RS-485 communication.

Note

- ***1) The automatic stability detection capability detects changes in pH value during calibration to determine whether the reading has become stable enough. When changes in pH value become smaller enough during a 10-second period following 25 seconds after the stability detection has been started, the stability level is determined as acceptable and then the value is held. When automatic stability detection is set to "no," the stability detection process automatically occurs, but the value is not held after the stability level has been determined as acceptable.**
- ***2) The automatic stability detection capability determines a simple increase or decrease as stable enough at 0.01 pH maximum for "HI", 0.02 pH maximum for "nor", or 0.03 pH maximum for "Lo."**

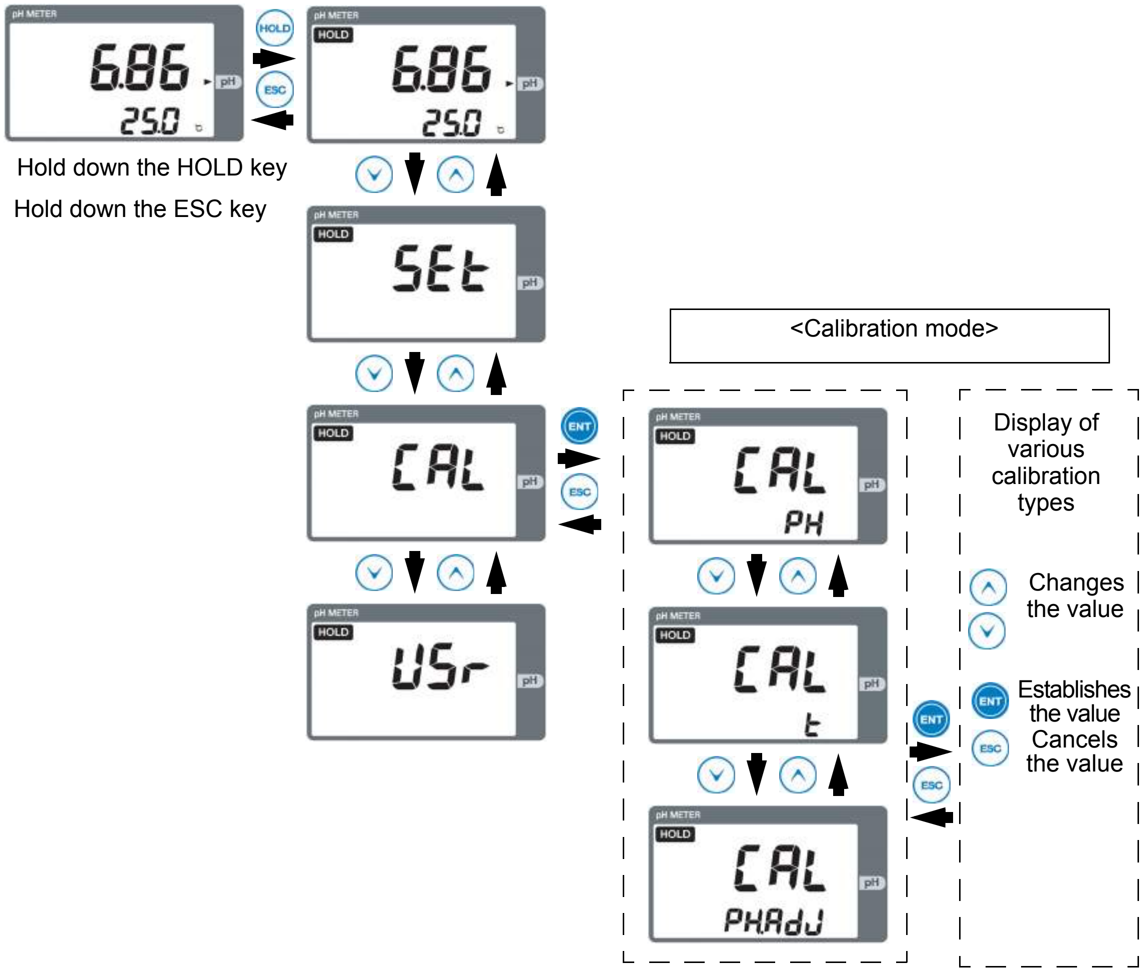
Calibration menu

Entering the Calibration menu allows you to calibrate the pH electrode or the temperature electrode, or to manually enter the calibration value for the pH electrode.

Entering the calibration menu

You can enter the calibration menu by either of two methods.
This section describes the method of entering this menu from the hold mode.

1. **In the meas mode, hold down the HOLD key until the HOLD indicator turns ON.**
2. **Press the [▲/▼] key to display "CAL" on the display part for the measured value.**
3. **Press the ENT key.**
4. **Press the [▲/▼] key to display a desired item on the auxiliary display part.**
5. **Press the [ENT] key.**



Tip

Holding down the CAL key in the meas mode allows you to directly go to the pH cal mode. For the normal calibration, use this operation to perform the calibration.

Table 3 Items displayed on the Cal menu

Screen 1	Screen 2	Calibration Type	Description	Action	Remarks
		"Auto"	First screen with the calibration type set to "Auto"	Immerse the pH electrode in the standard solution for pH7.	[ENT] key: Next screen
			Screen displayed during stability detection of standard solution	The measured value blinks and the left indicator changes incrementally.	After stability detection, this screen automatically changes over to the next screen.
			Screen displayed when the measurement of standard solution has been completed.	The measured value and all of the left indicators are illuminated. Immerse the pH electrode in the standard solution for pH4 or pH9.	● CAL key: Restarts stability detection ● ENT key: Goes to the next screen ● When the solution type is changed, this screen automatically changes over to the next screen.
			Screen displayed during stability detection of second-point standard solution	The measured value blinks and the left indicator changes incrementally.	● Use the standard solution for either pH4 or pH9 as the second one. ● After stability detection, this screen automatically changes over to the next screen.
			Screen displayed when the measurement of second standard solution has been completed	The measured value and all of the left indicators are illuminated.	This screen automatically changes over to the next screen.
			Screen showing that the calibration value has been established	When the calibration has been successfully completed, "good" blinks and the current mode automatically changes over to the hold mode.	Err (E-11 to E-15) is displayed when the calibration error occurs.

Screen 1	Screen 2	Calibration Type	Description	Action	Remarks
		"bASic" (one-point calibration)	First screen displayed for one-point calibration when the calibration type is "bASic"	Immerse the pH electrode in the standard solution.	● CAL key: Changes the number of calibration points ● [▲/▼] key: Changes a calibration point ● [ENT] key: Goes to the next screen
			Screen displayed during stability detection of standard solution	The measured value blinks and the left indicator changes incrementally.	
			Screen showing that the measurement of standard solution has been completed	The measured value and all of the left indicators are illuminated.	● CAL key: Restarts stability detection ● [▲/▼] key: Selects a solution type ● [ENT] key: Goes to the next screen
			Screen showing that the calibration value has been established	When the calibration is successfully completed, "good" blinks and the current mode automatically changes over to the hold mode.	Err (E-11 to E-15) is displayed when the calibration error occurs.

Screen 1	Screen 2	Calibration Type	Description	Action	Remarks
		"bASic" (2-point calibration)	First screen displayed for two-point calibration when calibration type is "bASic"	Immerse the pH electrode in the standard solution for the first point.	● CAL key: Changes the number of calibration points ● [▲/▼]: Changes the calibration point ● [ENT] key: Goes to the next screen
			Screen displayed during stability detection of standard solution	The measured value blinks and the left indicator changes incrementally.	
			Screen displayed when the measurement of standard solution has been completed	The measured value and all of the left indicators are illuminated. Immerse the pH electrode in the standard solution for the second point.	● CAL key: Restarts stability detection ● [▲/▼] key: Selects a solution type ● [ENT] key: Goes to the next screen
			Screen displayed during stability detection of second-point standard solution	The measured value blinks and the left indicator changes incrementally.	
			Screen displayed when the measurement of second-point standard solution has been completed	The measured value and all of the left indicators are illuminated.	● CAL key: Restarts stability detection ● [▲/▼] key: Selects a solution type ● [ENT] key: Goes to the next screen
			Screen displayed when the calibration value is established	When calibration is successfully completed, "good" blinks and the current mode automatically changes over to the hold mode.	Err (E-11 to E-15) is displayed when the calibration error occurs.
		"bASic" (3-point calibration)	First screen displayed for three-point calibration when calibration type is "bASic"	Immerse the pH electrode in the standard solution for pH4	● [CAL] key: Changes the number of calibration points ● [▲/▼] key: Changes the calibration point ● [ENT] key: Goes to the next screen
			Screen displayed during stability detection of standard solution	The measured value blinks and the left indicator changes incrementally.	
			Screen displayed when the measurement of standard solution has been completed	The measured value and all of the left indicators are illuminated. Immerse the pH electrode in the standard solution for the second point.	● [CAL] key: Restarts stability detection ● [▲/▼]key: Selects a solution type ● [ENT] key: Goes to the next screen
			Screen displayed during stability detection of second-point standard solution	The measured value blinks and the left indicator changes incrementally.	
			Screen displayed when the measurement of second-point standard solution has been completed	The measured value and all of the left indicators are illuminated. Immerse the pH electrode in the third standard solution.	● CAL: Restarts stability detection ● [▲/▼] key: Selects a solution type ● ENT key: Goes to the next screen

<table>
<tr><th>Screen 1</th><th>Screen 2</th><th>Calibration Type</th><th>Description</th><th>Action</th><th>Remarks</th></tr>
<tr><td rowspan="3"></td><td></td><td rowspan="3">"bASic" (three-point calibration)</td><td>Screen displayed during stability detection of third standard solution</td><td>The measured value blinks and the left indicator changes incrementally.</td><td></td></tr>
<tr><td></td><td>Screen displayed when the measurement of second-point standard solution has been completed</td><td>The measured value and all of the left indicators are illuminated.</td><td>● [CAL]: Restarts stability detection
● [▲/▼] key: Selects a solution type
● [ENT] key: Goes to the next screen</td></tr>
<tr><td></td><td>Screen displayed when the calibration value is established</td><td>When calibration is successfully completed, "good" blinks and the current mode automatically changes over to the hold mode.</td><td>Err (E-11 to E-15) is displayed when the calibration error occurs.</td></tr>
<tr><td></td><td></td><td>Temperature</td><td>Screen for temperature calibration</td><td>Calibrate the temperature value. Enter a variable amount and change the measured value to calibration temperature.</td><td>● [▲/▼] key: Specifies a temperature calibration value
● [ENT] key: Establishes the value
● Setting range: –5.0 to 5.0(°C)
Note: This setting is disabled when temperature sensor type is set to "non."</td></tr>
<tr><td rowspan="2"></td><td></td><td rowspan="2">Manual Input</td><td>Screen to enter the zero (STD) value for pH electrode</td><td>Enter the STD value obtained by calibration using another meter.</td><td>● [▲/▼] key: Specifies the STD value
● [ENT] key: Establishes the value and goes to the next screen
● Setting range: –90.0 to 90.0 mV
Initial value: 0.0 mV</td></tr>
<tr><td></td><td>Screen to enter the span (SLOPE) value for pH electrode</td><td>Enter the SLOPE value obtained by calibration using another meter</td><td>● [▲/▼]key: Specifies the SLOPE value
● [ENT] key: Establishes the value
● Setting range: 40.0 to 65.0 mV
Initial value: 59.1 mV</td></tr>
</table>

Note

When any value is entered on the PH.Adj screen, the previous calibration value is cleared. In entering a value, therefore, excersise care.

User check menu

Entering the User check menu allows you to check the display part or the transmission output or to initialize the settings.

● Entering the User check menu

1. **Hold down the HOLD key in the meas mode until the HOLD indicator turns ON.**
2. **Press the [▲/▼] key to display "USr" on the display part for the measured value.**
3. **Press the [ENT] key.**
4. **Press the [▲/▼] key to display a desired item on the auxiliary display part.**
5. **Press the [ENT] key.**

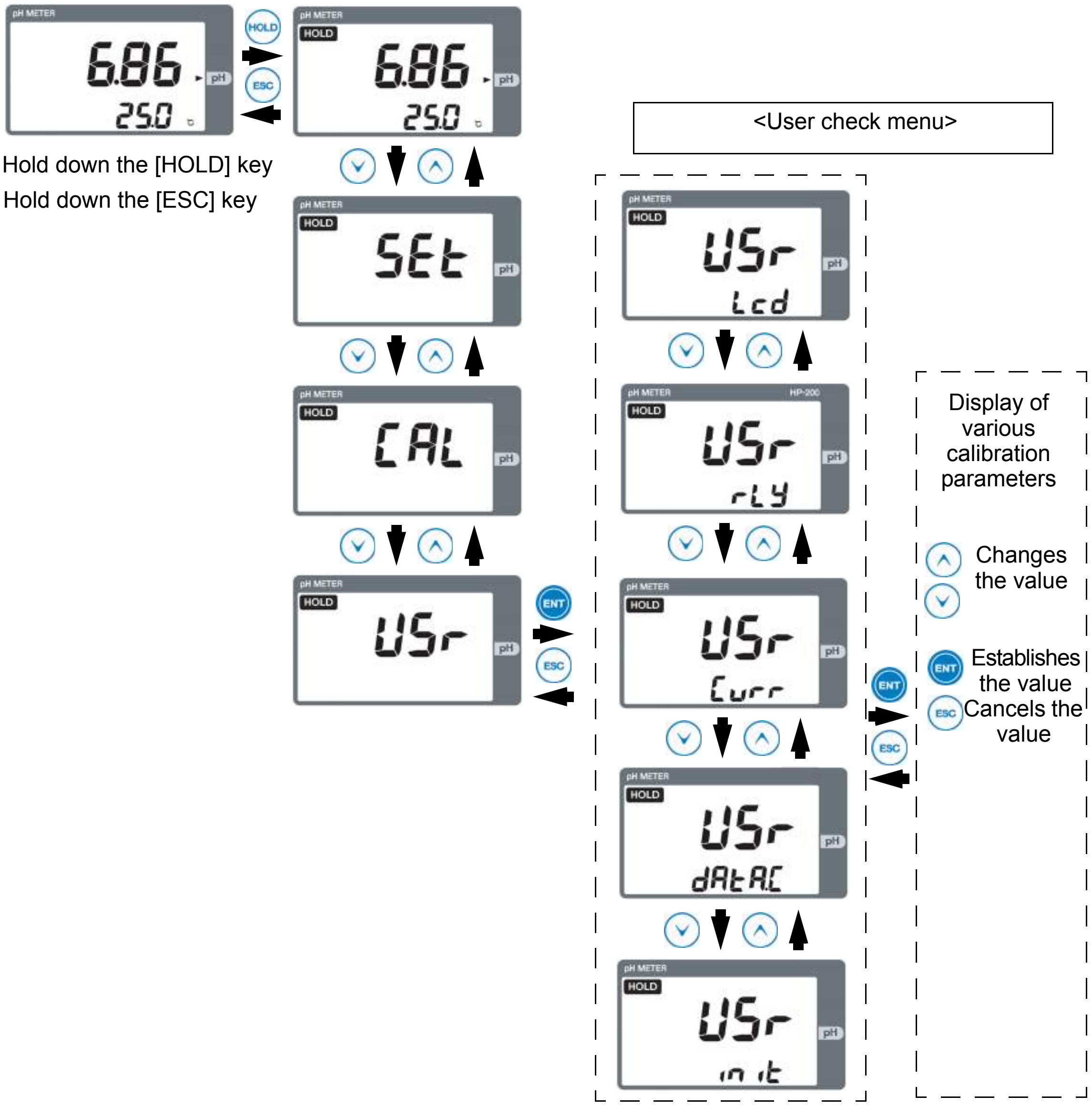
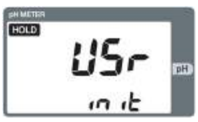


Table 4 Items displayed on the User check menu

Screen 1	Screen 2	Description	Action	Remarks
		Screen to check the LCD	The whole LCD is illuminated, allowing you to check the display.	
		Screen to check relay output 1	This screen allows you to check that relay 1 ("r1") turns ON/OFF	[ENT] key: Turns ON/OFF Note: When you enter this mode, contact output turns OFF.
		Screen to check relay output 2	This screen allows you to check that relay 2 ("r2") turns ON/OFF.	
		Screen to check relay output 3	This screen allows you to check that relay 3 ("r3") turns ON/OFF.	
		Screen to check relay output 4	This screen allows you to check that relay 4 ("r4") turns ON/OFF.	
		Screen to check FAIL output	This screen allows you to check that FAIL output turns ON/OFF.	
		Screen to check cleaning output	This screen allows you to check that cleaning ("CLn") turns ON/OFF.	

Screen 1	Screen 2	Description	Action	Remarks
		Screen to check transmission output (to check output of 4 mA)	4 mA is output to transmission output 1 and 2.	● [▲] key: Allows you to check output of 12 mA ● [▼] key: Allows you to check output of 3.8 mA
		Screen to check transmission output (to check output of 12 mA)	12 mA is output to transmission output 1 and 2.	● [▲] key: Allows you to check output of 20 mA ● [▼] key: Allows you to check output of 4 mA
		Screen to check transmission output (to check output of 20 mA)	20 mA is output to transmission output 1 and 2.	● [▲] key: Allows you to check output of 21 mA ● [▼] key: Allows you to check output of 12 mA
		Screen to check transmission output (to check output of 21 mA)	21 mA is output to transmission output 1 and 2.	[▲] key: Allows you to check output of 20 mA
		Screen to check transmission output (to check output of 3.8 mA)	3.8 mA is output to transmission output 1 and 2.	[▼] key: Allows you to check output of 4 mA
		Screen to check data: pH and temperature	This screen allows you to check pH and temperature.	The measured temperature value is displayed even if temperature electrode is set to "non."
		Screen to check data: pH value and pH electrode voltage	This screen allows you to check the pH value and voltage of the pH electrode.	
		Screen to check data: pH value and temperature sensor type	This screen allows you to check the pH value of the pH electrode and the temperature sensor type (500 Ω, 1 kΩ, 6.8 kΩ, or 10 kΩ).	The automatically detected temperature sensor type is displayed.
		Screen to initialize the settings and the calibration data	This screen allows you to initialize the settings and the calibration data. When "yES" is selected, they are initialized.	● Exercise care as the settings and the calibration values are reset to the initial values. ● Upon completion of initialization, the initial screen is displayed with the meas mode selected.

Calibration

Calibration is classified into pH calibration and temperature calibration.

Note

After the power has been first turned ON or the electrode has been replaced, be sure to perform calibration with a standard solution.

pH calibration

pH calibration is classified into three kinds: auto, basic, and manual.

Reference

For selecting auto or basic calibration, see " Set calibration type " (page 33).

Tip

Hold down the CAL key in the meas mode. This allows you to directly display the pH cal mode (for auto or basic calibration only).

Note

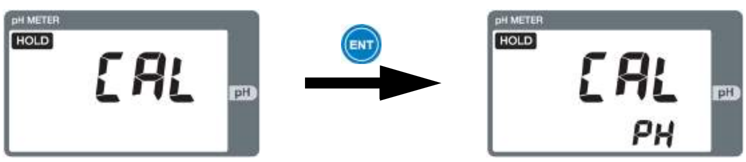
- **Do not reuse any standard solution.**
- **When you enter the hold mode, transmission output becomes the output value specified on the Setup menu.**
  [See" Set currently held value for transmission output 1 " (page 36) " Set currently held value for transmission output 2 " (page 37)]
- **When temperature sensor type "SEnSor" is set to "non" on the Setup menu, temperature is assumed to be 25°C regardless of the solution temperature setting for "SoL.t.**

Operation procedure

Enter the pH cal mode from the meas mode.

1. **Hold down the HOLD indicator in the meas mode until the HOLD indicator turns ON.**
   The hold mode is selected with HLD displayed on the upper left part.

2. **Press the ▼ key to display "CAL" on the display part for the measured value and then press the ENT key.**
   "pH" is displayed on the auxiliary display part.



3. **Press the ENT key.**
   The pH cal mode is selected with the Auto or Basic cal screen displayed.
   Carry out calibration in accordance with the operation procedure for auto or basic calibration described on the next and subsequent pages.

Tip

- **To cancel the calibration during its process, press the ESC key. You will return to the Cal menu or the hold mode without changing calibration data.**
- **If an error occurs, e.g., from immersing the electrode in a solution of any pH value other than the corresponding setting on the Setup menu, take remedial action as instructed with an error code.**

Auto calibration

Calibrate the first point (pH7) and then the second point (pH4 or pH9 standard solution).

Items to be supplied

- Standard solution for pH7
- Standard solution for pH4 or pH9

Operation procedure

Calibrate the first point.

Tip

- **To cancel the calibration during its process, press the ESC key.**
- **If en error is displayed, follow the instructions given with the error code.**

1. **Perform the operation shown in“ pH calibration ” (page 47) to select the pH cal mode.**

2. **Immerse the electrode in the standard solution for pH7 and then press the ENT key.**
   The first point will be started with the measured value blinking on the display part for the measured value.
   Once the stability is determined as acceptable, the measured value stops blinking and the first-point calibration is finished.

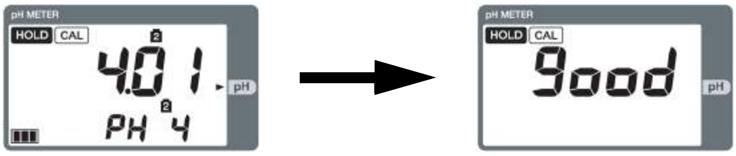
Subsequently, perform the second-point calibration.

Tip

To perform the calibration again with the same solution, press the CAL key in this state.
The calibration for pH7 will be started.

3. **To perform the second-point calibration, immerse the electrode in the standard solution for pH4 or pH9 and then wait in this state or press the ENT key.**
   The second-point calibration will be started.

4. **When the reading becomes stable enough, the second-point calibration is finished.**
   Once the stability is determined as acceptable, the measured value stops blinking and the second-point calibration is finished.
   Subsequently, the calibration value is updated with "good" blinking.



At this point, the new calibration value is applied.
Now the pH auto calibration is complete with the Cal menu or the Hold menu displayed again.

When calibration is performed from the Cal menu

When calibration is performed by holding down the CAL key to enter the pH cal mode

Tip

To return to the meas mode, hold down the HOLD key or the ESC key.
This will cancel the hold mode, bringing you back to the meas mode.

Basic calibration

Use this calibration when you want to perform any calibration method other than the two-point calibration for pH4 and pH7 or for pH7 and pH9. It also allows you to achieve a known pH value.

Items to be supplied
- 1 to 3 kinds of standard solutions
- Two or three kinds of standard solutions that have a difference of 2 pH minimum between them when two- or three-point calibration is performed.

Operation procedure

Tip
- **To cancel calibration during its process, press the ESC key.**
- **If an error occurs, follow the instructions given with the error code.**

1. Perform the operation shown in“ pH calibration ” (page 47) to select the pH cal mode.

For 1-point calibration

For 2-point calibration

For 3-point calibration

2. Press the [▲/▼] key to change the kind of standard solution used.
Pressing the CAL key allows you to change the number of calibration points.

Tip
- **For only the one- and two-point calibration, an arbitrary standard solution may be used. When calibration is performed using an arbitrary standard solution, display "USr" on the auxiliary display part.**
- **For the three-point calibration, be sure to use the standard solution for pH7 for the first point.**

When the number of calibration points and standard solutions are determined, immerse the electrode in the standard solution for the first point.

3. Press the ENT key to start the first-point calibration.
The stability detection of standard solution for the first point is started.

For 1-point calibration

For 2-point calibration

For 3-point calibration

The measured value blinks and the indicator changes incrementally. When the reading becomes stable enough, the standard solution is automatically detected and then the closest kind of standard solution is displayed on the auxiliary display part.

Tip
- **If the kind of standard solution displayed on the auxiliary display part is different, it may be changed using the [▲/▼] key. The changeable range is determined based on the signal from the electrode. For the three-point calibration, however, the kind of standard solution for the first point cannot be changed.**
- **Press the CAL key. Calibration will be started using the same solution**
- **When the standard solution is "USr," use the [▲/▼] key to select an arbitrary standard solution value.**

4. Press the ENT key.

The auxiliary display part will blink and the calibration of the first point will be finished. (Now the one-point calibration is complete.)

Tip

The one-point calibration updates only the zero (STD) value.

For the calibration of two points or more, immerse the electrode in the standard solution for the second points.

The stability detection of standard solution for the second point will be started.

For 1-point calibration

For 2-point calibration

For 3-point calibration

The measured value blinks and the indicator changes incrementally. When the reading becomes stable enough, the standard solution is automatically detected and the closest kind of standard solution is displayed on the auxiliary display part while blinking.

Tip

- **If the kind of standard solution displayed on the auxiliary display part is different, it can be changed using the [▲/▼] key. The changeable range is determined based on the signal from the electrode.**
- **Pressing the CAL key restarts the calibration using the same solution.**
- **When the standard solution is "USr," use the [▲/▼] key to change it to an arbitrary standard solution after the completion of calibration.**

5. Press the ENT key.

The auxiliary display part will blinks and the calibration of the second point will be finished. (Now the two-point calibration is complete.)

For the three-point calibration, immerse the electrode in the standard solution for the third point.

The stability detection of the standard solution for the third point will be started.

For 2-point calibration

For 3-point calibration

The measure value blinks and the indicator changes incrementally. When the reading becomes stable enough, the standard solution is automatically detected and the closest kind of standard solution is displayed on the auxiliary display part.

Tip

- **If the kind of standard solution displayed on the auxiliary display part is different, it can be changed using the [▲/▼] key. The changeable range is determined based on the signal from the electrode.**
- **Pressing the CAL key restarts the calibration using the same solution.**

6. Press the ENT key.

The auxiliary display part will blink and the calibration of the third point will be finished. Now the three-point calibration is complete.

For 3-point calibration

At this point, the new calibration value is applied.
Now the pH basic calibration is complete. The Cal menu is displayed or the hold mode becomes active.

When calibration is performed from the Cal menu

When calibration is performed by holding down the CAL key to enter the pH cal mode

Tip

To return to the meas mode, hold down the HOLD key or the ESC key.
The hold mode will be canceled and you will return to the meas mode.

Temperature calibration

The temperature calibration is used to correct temperature.

Note

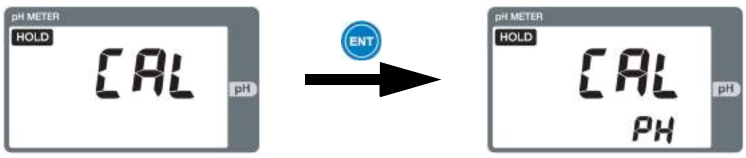
- **Immerse the electrode in a solution at a known temperature and then well expose it to the water temperature.**
- **When you enter the hold mode, transmission output becomes the output value specified on the Setup menu.**
  (See“ Set currently held value for transmission output 1 ” (page 36)“ Set currently held value for transmission output 2 ” (page 37))
- **When temperature sensor type "SEnSEr" is set to "non" on the Setup menu, you cannot enter the temp cal mode.**

Operation procedure

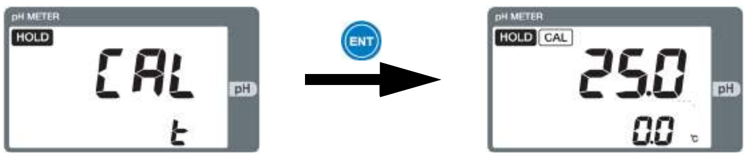
Enter the temp cal mode from the meas mode.

1. **Hold down the HOLD key in the meas mode until the HOLD indicator turns ON.**
The hold mode will becomes active with HOLD displayed on the upper left part.

2. **Press the [▼] key to display "CAL" on the display part for the measured value and then press the ENT key.**
"PH" will be displayed on the auxiliary display part.



3. **Press the [▼] key to display "t" on the auxiliary display part and then press the ENT key.**
The temp cal mode will become active and the auxiliary display part will start blinking.



4. **Press the [▲/▼] key to adjust the value displayed on the display part for the measured value so that it becomes the water temperature.**
[▲] key: Changes the value in increments of 0.1°C
[▼] key: Changes the value in decrements of 0.1°C
The setting range is between -5.0°C and 5.0°C.

5. **Press the ENT key.**
The temperature calibration coefficient will be updated.
Now the temperature calibration is complete with the Cal menu displayed.

Tip

- **To cancel the calibration during its process, press the ESC key.**
  You will return to the Cal menu and the calibration data will not be updated.
- **To subsequently set any other parameters for calibration, press the [▲/▼] key to select a parameter.**
- **To return to the meas mode, hold down the HOLD key or the ESC key.**
  The hold mode will be canceled and you will return to the meas mode.

pH calibration by manual input

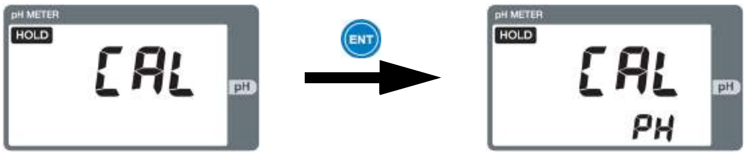
For this calibration, manually enter the calibration value for pH.

Operation procedure

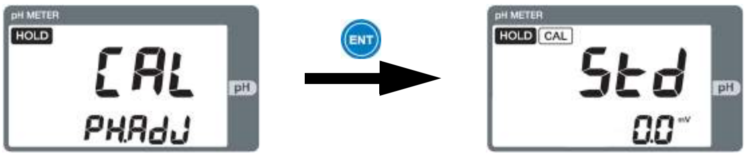
Enter the manual input pH cal mode from the meas mode.

1. **Hold down the HOLD key in the meas mode until the HOLD indicator turns ON.**
   The hold mode will become active with HOLD displayed on the upper left part.
   The transmission output stops with the measured value held.

2. **Press the [▼] key to display "CAL" on the display part for the measured value and then press the ENT key.**
   "PH" will be displayed on the auxiliary display part.

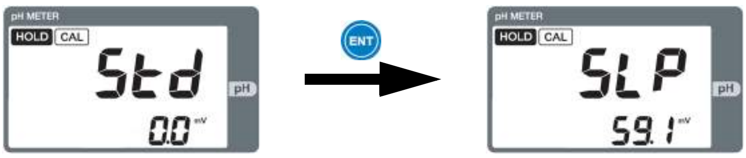


3. **Press the [▼] key to display "PH.Adj" on the auxiliary display part and then press the ENT key.**
   The STD value input ??? screen will be displayed in the pH manu input cal mode.



4. **Press the [▲/▼] key to enter the STD value.**
   [▲] key: Changes the value in increments of 0.1 mV
   [▼] key: Changes the value in decrements of 0.1 mV
   The setting range is between -90.0 mV and 90.0 mV.

5. **Press the ENT key.**
   The STD value will be established. The SLOPE input screen will be displayed and the auxiliary display part will start blinking.



6. **Press the [▲/▼] key to enter the SLOPE value.**
   [▲] key: Changes the value in increments of 0.1 mV/pH
   [▼] key: Changes the value in decrements of 0.1 mV/pH
   The setting range is between 40.0 mV/pH and 65.0 mV/pH.

7. **Press the ENT key.**
   The SLOPE value will be updated.
   Now the pH calibration by manual input is complete with the Cal menu displayed.

Tip

- **To cancel the calibration during its process, press the ESC key.**
  You will return to the Cal menu with no calibration data updated.
- **To subsequently set any other calibration parameter, press the [▲/▼] key to select the parameter to be set.**
- **To return to the meas mode, hold down the HOLD key or the ESC key.**
  The hold mode will be canceled and you will return to the meas mode.
- **When temperature sensor type "SEnSor" is set to "non" on the Setup menu, the calibration is performed assuming a temperature of 25°C regardless of the setting "SoL.t" for solution temperature.**

Contact output

Specify a control action for each of R1, R2, R3, and R4. The available control items vary depending on the selected output type. The items marked with a circle are available for display as options.

Screen	Description	Selected type							
		"non"	"CtrL"	"AL"	"tPS"	"tPF"	"t"	"HoLd"	"CLn"
	Setting for contact output type ("r1" or "r2")	○	○	○	○	○	○		
	Setting for contact output type ("r3" or "r4)	○	○	○			○	○	○
	Setting for control method		○	○	○	○	○		
	Setting for control valve		○	○	○	○	○		
	Setting for control width		○						
	Setting for contact output delay time			○			○	○	
	Setting for proportional band				○	○			
	Setting for period				○	○			
	Setting for control shift amount				○				
	Setting for minimum ON time				○	○			
	Setting for maximum control amount				○	○			
	Setting for F-zone amount					○			
	Setting for maximum F-zone extension time					○			
	Setting for cleaning output type								○

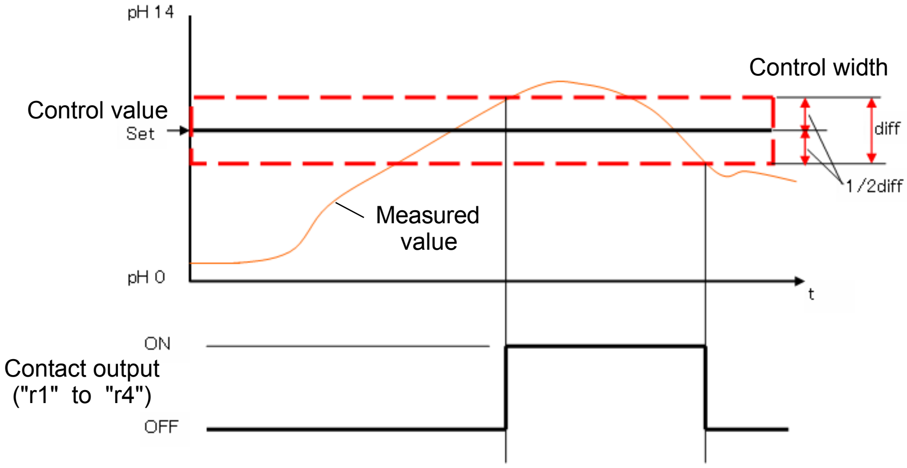
Contact output types

The following contact output types are available:

- "non": No contact output
  This setting provides no contact output.

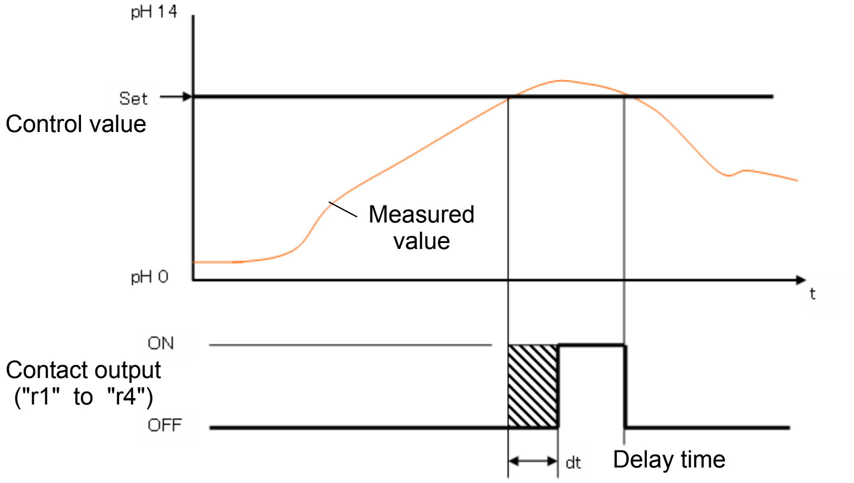
- "CtrL": Control output
  When the measured value is larger than (control value plus control width x 1/2), the control output is turned ON to drive the metering pump so that the chemical agent is injected. When the measured value is smaller than (control value minus control width x 1/2), the control output is turned OFF to stop driving the pump so that the injection of chemical agent is stopped.
  The available settings are shown below:
  - "HorL": Setting for control method
    "Hi": The control output is turned ON when the measured value becomes larger than (control value plus control width x 1/2).
    "Lo": The control output is turned ON when the measured value becomes smaller than (control value minus control width x 1/2).
  - "SEt": Setting for control value (0.00 to 14.00 pH)
  - "diff": Setting for control width (0.02 to 4.00 pH)

"CtrL": Example of control output (when the control method is "Hi"); reverse this example when the control method is "Lo"



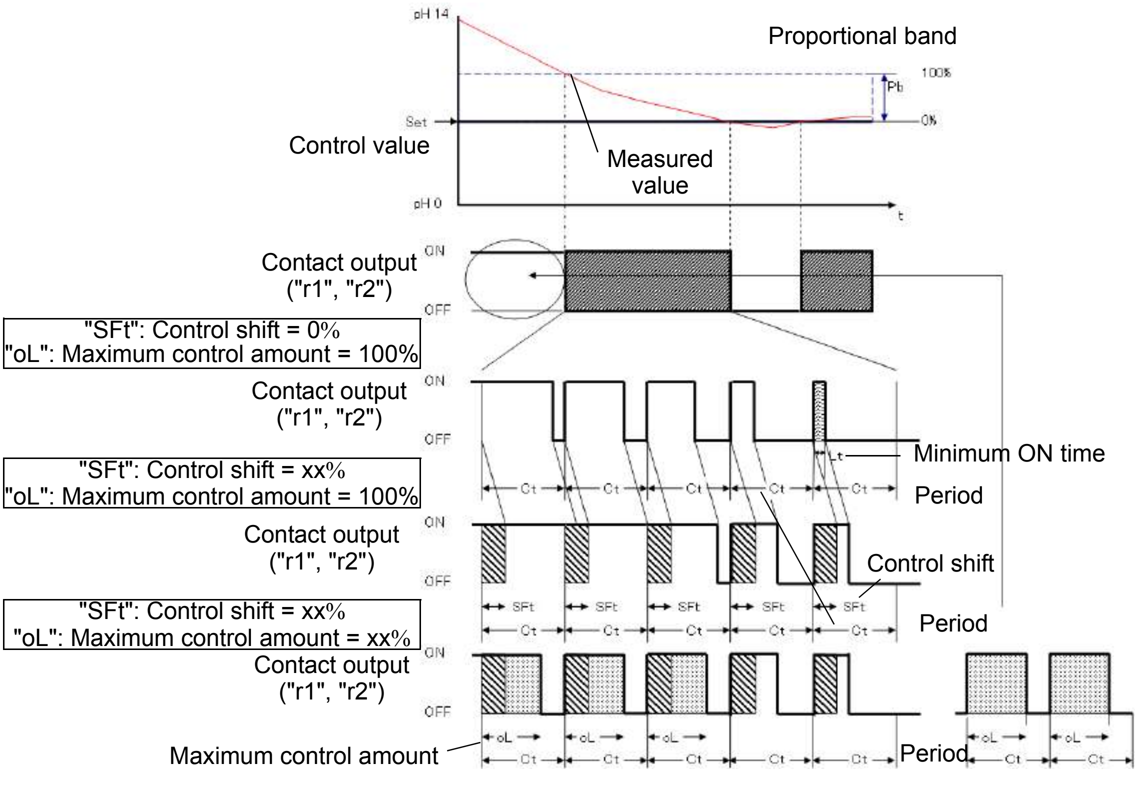
- "AL": Alarm output
  When the measured value becomes larger than the setting, the alarm output is turned ON to trigger the alarm after the delay time. When the measured value becomes smaller than the setting, the output is immediately turned OFF to cancel the alarm. (The above case is applicable for the upper-limit operation; the reversed case is applicable for the lower-limit operation.)
  The available settings are shown below:
  - "HorL": Setting for control method
    "Hi": The control output is turned ON when the measured value becomes larger than the upper limit.
    "Lo": The control output is turned ON when the measured value becomes smaller than the lower limit.
  - "SEt": Setting for control value (0.00 pH to 14.00 pH)
  - "dt": Setting for delay time (0 to 600 seconds)

"AL": Example of alarm output (when the control method is "Hi"; the reversed example is applicable for "Lo")



- "tPS": Time-sharing proportional control (shift function)
  Time-sharing proportional control means variably controlling the output by changing the ON/OFF proportion within a certain period in accordance with the deviation (difference between control value and measured value) in order to perform proportional control for a metering pump for pH control or a control device such as a solenoid valve.
  Time-sharing proportional control (shift function) is an effective method when the control result does not reach the control, e.g., in a continuous treatment process.
  When a shift is applied as shown below, the ON time becomes longer even if the deviation remains the same:
  The available settings are shown below:
  - "HorL": Setting for control method
    "Hi": The control output is turned ON when the measured value becomes larger than the upper limit.
    "Lo": The control output is turned ON when the measured value becomes smaller than the lower limit.
  - "SEt": Setting for control value (0.00 pH to 14.00 pH)
  - "pb": Setting for proportional band (0.02 pH to 4.00 pH)
  - "Ct": Setting for period (5 to 300 seconds)
  - "SFt": Setting for control shift amount (0% to 50%)
  - "Lt": Setting for minimum ON time (2 to 10 seconds)
  - "oL": Setting for maximum control amount (50% to 100%)

"tPS": Example of time-sharing proportional control output (shift function);
the reversed example is applicable for "Lo."



Tip

When control method is set to "Hi," ON/OFF time is as shown below:

t_{on} = Ct x [(measured vaue - SEt) / Pb + SFt/100]

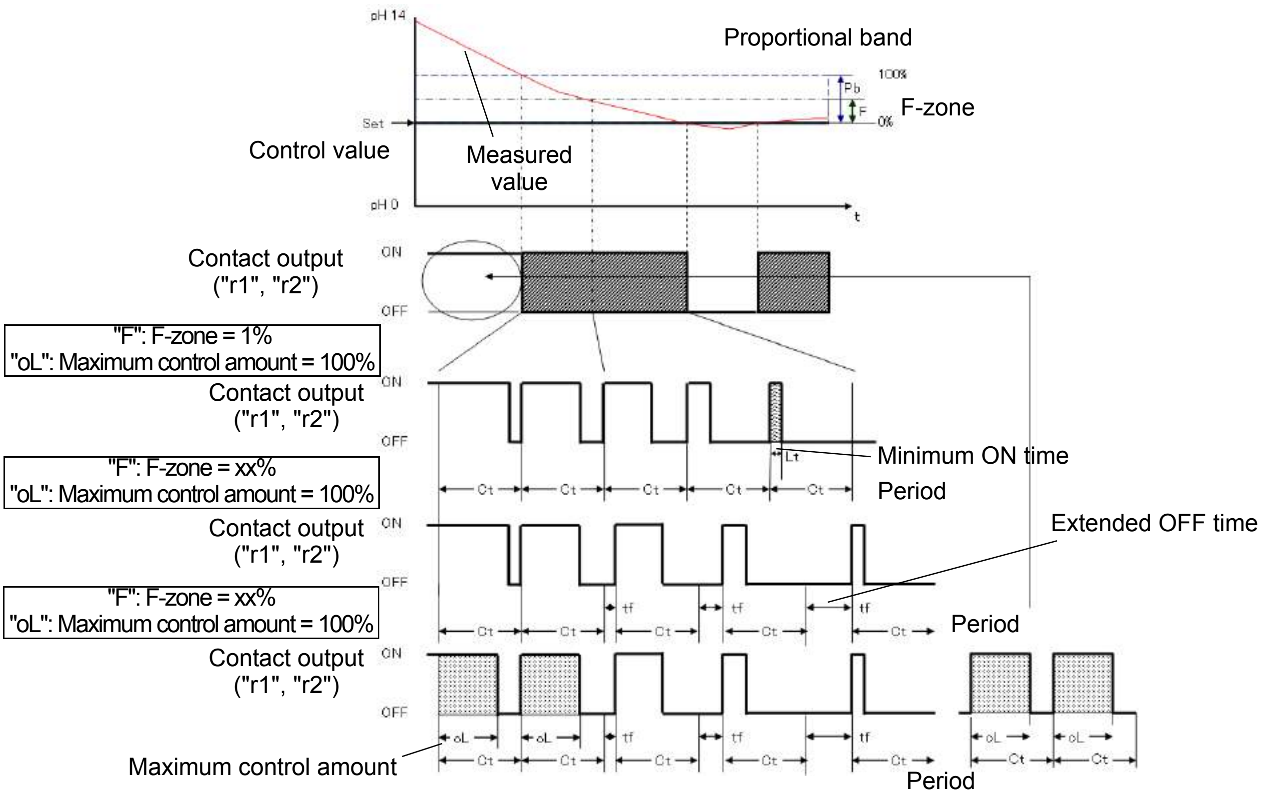
When t_{on} is smaller than Lt, t_{on} = Lt; when t_{on} is larger than (Ct x oL/100), t_{on} = (Ct x oL/100); and when t_{on} is larger than Ct, t_{on} = Ct.

t_{off}=Ct-t_{on}

t_{on}: ON time in seconds, t_{off}: OFF time in seconds

- "tPF": Time-sharing proportional control (F-zone function)
  Time-sharing proportional control means variably controlling the output by changing the ON/OFF proportion within a certain period in accordance with the deviation (difference between control value and measured value) in order to perform proportional control for a metering pump for pH control or a control device such as a solenoid valve.
  pFor pH control, it is very important to control the chemical agent when the deviation becomes smaller. Because of a characteristic specific to pH control, the injection of a small amount of chemical agent may cause the control result to exceed the control value. Time-sharing proportional method (F-zone function) has been devised as a method to fine-tune the ON/OFF proportion by extending the OFF time for the normal time-sharing proportional control when the deviation becomes smaller. It is an effective control method for a batch treatment process.
  This is an effective control method for cases where the control result greatly exceeds the control value, e.g., in a batch treatment process.
  The available settings are shown below:
  - "HorL": Setting for control method
    "Hi": The control output is turned ON when the measured value becomes larger than the upper limit.
    "Lo": The control output is turned ON when the measured value becomes smaller than the lower limit.
  - "SEt": Setting for control value (0.00 pH to 14.00 pH)
  - "pb": Setting for proportional band (0.02 pH to 4.00 pH)
  - "Ct": Setting for period (5 to 300 seconds)
  - "Lt": Setting for minimum ON time (2 to 10 seconds)
  - "oL": Setting for maximum control amount (50% to 100%)
  - "F": Setting for F-zone (1% to 100%)
  - "ctE": Setting for period extension time (0 to 300 seconds)

"tPS": Example of time-sharing proportional control output (F-zone function);
the reversed example is applicable for "Lo."



Tip

- **When control method is set to "Hi," the ON/OFF time is calculated using the equation below:**
  t_{on} = Ct x (measured value - SEt) / Pb
  When t_{on} is smaller than Lt, t_{on} = Lt; when t_{on} is larger than (Ct x oL/100), t_{on} = (Ct x oL/100); and when t_{on} is larger than Ct, t_{on} = Ct.
  t_{off}=Ct-t_{on}
  t_{on}: ON time in seconds t_{off}: OFF time in seconds
- **The extended OFF time "tf" is calculated using the equation below:**
  t_f = ctE x [(F/100 - (measured value - SEt) / Pb) / (F/100)]^2
  t_f: Extended OFF time in seconds

- "t": Temperature alarm output
  When the temperature value is higher than the setting, this output is turned ON to trigger the alarm after the delay time. When the temperature value becomes lower than the setting, the output is immediately turned OFF to cancel the alarm. The above case is applicable for the upper-limit operation; the reversed case is applicable for the lower-limit operation.
  The available settings are shown below:
  - "HorL": Setting for control method
    "Hi": The alarm output is turned ON when the temperature value becomes larger than the upper limit.
    "Lo": The alarm output is turned ON when the temperature value becomes smaller than the lower limit.
  - "SEt": Setting for control value (-10.0°C to 120.0°C)
  - "dt": Setting for delay time (0 to 600 seconds)

- "HoLd: Output during hold mode
  When the measured value is held, this output is turned ON after the delay time. When the hold mode is canceled, the output is immediately turned OFF.
  The available settings are shown below:
  - "dt": Setting for delay time (0 to 600 seconds)

- "CLn": Output during cleaning mode
  This is the contact output during cleaning.
  The available settings are shown below:
  - "CLn.t": Setting for cleaning output type
    "CLn": Hold time during and after cleaning; contact output
    "trg": 5-second contact output 2 seconds after cleaning has been finished

Tip

"CLn": When output during cleaning is set to "trg," sending this output to the contact input on our meter of the same series enables the chained operation of cleaning, i.e., the continuous cleaning operation that is performed at time delayed by each meter. The use of this operation eliminates the need of considering any pressure decrease in the water or air source for water or air cleaning.

Cleaning output

The selected cleaning output (CLN output) is provided under the following conditions

With or without cleaning output "CLn"	Cleaning output type "typE"	Internal cleaning period (during cleaning time: ○, during wait time: -)	External contact input (ON: ○, OFF: -)	Cleaning output*1 (ON: ⊙, OFF: -)	"CLn" contact output*2 (ON: ●, OFF: -)	"HoLd" contact output*3 (ON: ●, ○ OFF: -)	Transmission output*2
"yES"	"nor"	○	○	⊙	●	●	*4
		○	-	⊙	●	●	
		-	○	-	-	○	
		-	-	-	-	-	Measured value
	"or"	○	○	⊙	●	●	*4
		○	-	⊙	●	●	
		-	○	⊙	●	●	
		-	-	-	-	-	Measured value
	"And"	○	○	⊙	●	●	*4
		○	-	-	-	-	Measured value
		-	○	-	-	-	
		-	-	-	-	-	
	"trg"	No setting	ON for 2 seconds	⊙	●	●	*4
			-	-	-	-	Measured value
"no"	No setting	No setting	○		-	○	*4
			-		-	-	Measured value

Note

- ***1: The cleaning output is provided in accordance with " Set cleaning output time (when cleaning output is set to "yES") " (page 38).**
- ***2: The "Cln" contact output is provided during cleaning output and during the set hold time after cleaning output.**
- ***3: The "HoLd" contact output is provided either during cleaning output and during the set hold time after cleaning output (●) or when the external contact input is ON (○).**
- ***4: The transmission output is provided in accordance with " Set currently held value for transmission output 1 " (page 36), " Set currently held value for transmission output 2 " (page 37).**

Output conditions for various output types

The output conditions for various output types are shown below:

<table>
<tr><th colspan="3" rowspan="3">Status</th><th rowspan="3">Contact output r1 to r4</th><th rowspan="3">Cleaning output CLN</th><th colspan="6">Fail output FAIL</th><th colspan="4">Transmission output</th></tr>
<tr><th colspan="2">Self-check When error occurs “F.SC“</th><th colspan="2">When measured value deflects from scale "F.oF"</th><th colspan="2">When temperature is abnormal "F.tE"</th><th colspan="2">Transmission output 1</th><th colspan="2">Transmission output 2</th></tr>
<tr><th>yES</th><th>no</th><th>yES</th><th>no</th><th>yES</th><th>no</th><th>Without burnout</th><th>With burnout</th><th>Without burnout</th><th>With burnout</th></tr>
<tr><td rowspan="9">Measurement</td><td colspan="2">When normal</td><td>Enabled</td><td rowspan="9">Enabled</td><td colspan="6">OFF</td><td colspan="4">Measurement</td></tr>
<tr><td colspan="2">When measured value deflects from scale</td><td>EnabledEnabled (only "F.oF" is ON when held value is specified)</td><td colspan="2">OFF</td><td>ON</td><td colspan="3">OFF</td><td>Measured value (only when FAIL signal output is ON, the directly previous value or the preset value is output)</td><td>Measured value (only when FAIL signal output is ON, burnout output occurs)</td><td>Measured value (only when FAIL signal output is ON, the directly previous value or the preset value is output)</td><td>Measured value (only when FAIL signal output is ON, the directly previous value or the preset value is output)</td></tr>
<tr><td rowspan="3">When error occurs</td><td>E-21 to E-25</td><td rowspan="3">OFF (ON when hold mode is selected)</td><td colspan="4">OFF</td><td>ON</td><td>OFF</td><td rowspan="3">Directly previous value or preset value</td><td rowspan="2">Directly previous value or preset value (only when FAIL signal output is ON, burnout output occurs)</td><td rowspan="3">Directly previous value or preset value</td><td>Directly previous value or preset value (only when FAIL signal output is ON, burnout output occurs)</td></tr>
<tr><td>E-71, E-72</td><td>ON</td><td colspan="5">OFF</td><td>Directly previous value or preset value</td></tr>
<tr><td>E-90 to E-92</td><td colspan="6">ON</td><td>Burnout output</td><td>Burnout output</td></tr>
<tr><td colspan="2">During external input holding or during cleaning</td><td>OFF (ON when hold mode is selected; ON during cleaning when Cln mode is selected)</td><td colspan="6">OFF</td><td colspan="4" rowspan="4">Directly previous value or preset value</td></tr>
<tr><td rowspan="3"></td><td>E-21 to E-25</td><td rowspan="3">OFF (ON when hold mode is selected)</td><td colspan="6">OFF</td></tr>
<tr><td>E-71, E-72</td><td colspan="6">OFF</td></tr>
<tr><td>E-90 to E-92</td><td colspan="6">ON</td></tr>
<tr><td colspan="3">Hold mode (including calibration)</td><td rowspan="2">OFF (ON when hold mode is selected)</td><td rowspan="2">OFF</td><td colspan="6">OFF</td><td colspan="4" rowspan="2">Directly previous value or preset value, or measured value (only in the cal mode)</td></tr>
<tr><td></td><td>When error occurs</td><td>E-90 to E-92</td><td colspan="6">ON</td></tr>
</table>

Information for accurate measurements

This chapter provides information on maintenance in order to ensure the highest accuracy and best use of the HP-200.
It describes accuracy maintenance including the daily calibration.

Daily calibration (cal mode)

The method of daily performing calibration is described below:

Note

- **Do not reuse any standard solution.**
- **The transmission output becomes the state specified in and “ Set currently held value for transmission output 1 ” (page 36) “ Set currently held value for transmission output 2 ” (page 37).**

Calibration operations

Perform these operations in the meas mode.

1. **Hold down the CAL key until the CAL indicator turns ON.**
   The pH cal mode will become active and the calibration type (auto or basic calibration) selected in the Setup menu will be displayed.

2. **Perform calibration by the method described in “ Auto calibration ” (page 48) or “ Basic calibration ” (page 50).**

Tip

- **To cancel the calibration during its process, press the ESC key. You will be brought back to the hold mode without updating the calibration data.**
- **If an error occurs, e.g., from immersing the electrode in a solution of a different pH value from the one specified in the basic settings, take remedial action in accordance with the instructions given by the error code.**
- **To perform the temperature calibration, enter the hold mode.**

Maintenance procedure for HP-200

Checks of HP-200

The decreased insulation of the electrode connector makes it difficult to make accurate measurements.
Check the electrode plate periodically (about once every year) to see whether it is free from rust and any other cause for the decreased insulation.
Use soft cloth to wipe off dirt on the case for the HP-200.

Note

Do not use any organic solvent.

Maintenance of electrode

This section describes the maintenance method of a general pH electrode.
For further details, refer to the instruction manual for each electrode.

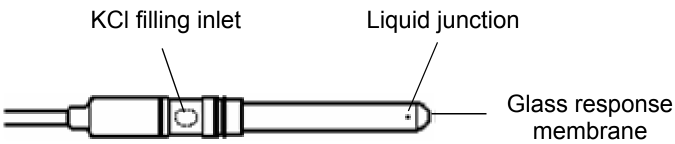
Cleaning the electrode

Dirt on the electrode can delay the response speed, reading drift, and instability. Periodically check the tip (glass membrane) and the liquid junction of the electrode and maintain them clean by rinsing off any dirt and coating.
It is recommended that the electrode be cleaned when calibration is performed using a standard solution. In this case, clean the electrode before the calibration.

Note

When a cleaner is installed, stop its operation and then perform the maintenance of the electrode. Turn OFF the HP-200 or hold down the HOLD key to enter the hold mode and then start the maintenance.

View of pH electrode alone



Cleaning the pH electrode

[1], [2], and [3] indicate the cleaning steps for different dirt levels.
If the characteristics cannot be restored by performing step [1], go on to step [2] or even step [3]. If the characteristics are still not restorable, replace the electrode as its service life has expired.

⚠ WARNING

Strong acid hazard

If dilute hydrochloric acid gets into your eye, its mucosa may be irritated, leading to blindness.

In handling hydrochloric acid, be sure to wear protective goggles, protective gloves, and a protective mask.
If dilute hydrochloric acid gets into your eye, immediately rinse the eye with a large quantity of water for more than 15 minutes. In rinsing the eye, fully open the lid with your fingers so that the eyeball and the lid are well exposed to running water.
If dilute hydrochloric acid adheres to your body and clothes, burns may result. Therefore, immediately take off the clothes and then rinse the body with a large quantity of water.

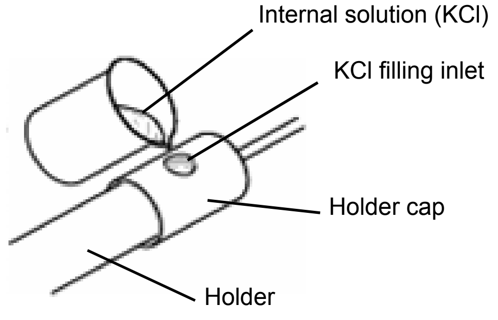
Note

Do not immerse the electrode in dilute hydrochloric acid for long hours.

	General dirt	Removal of soft foreign matter Organic matter Fiber Algae	Removal of adhesive foreign matter Oil Organic matter	Removal of hard foreign matter Calcium salt Inorganic salt
☉Common work Rinse with deionized water and wipe off dirt with gauze	[1]	[1]	[1]	[1]
Wipe off with gauze soaked with organic solvent and then rinse with deionized water	[2]	[2]	[2]	-
Wipe off with gauze soaked with detergent and then rinse with deionized water	-	[3]	[3]	-
Immerse in dilute hydrochloric acid (1 mol/L) for 15 seconds and then rinse with deionized water (repeat this process)	[3]	-	-	[2]

Adding KCl internal solution

For accurate measurements, the level of the internal solution (3.33 mol/L KCl solution) in the electrode must be higher by 10 cm minimum than the level of the solution under measurement. Periodically add the internal solution so that this level can be maintained.

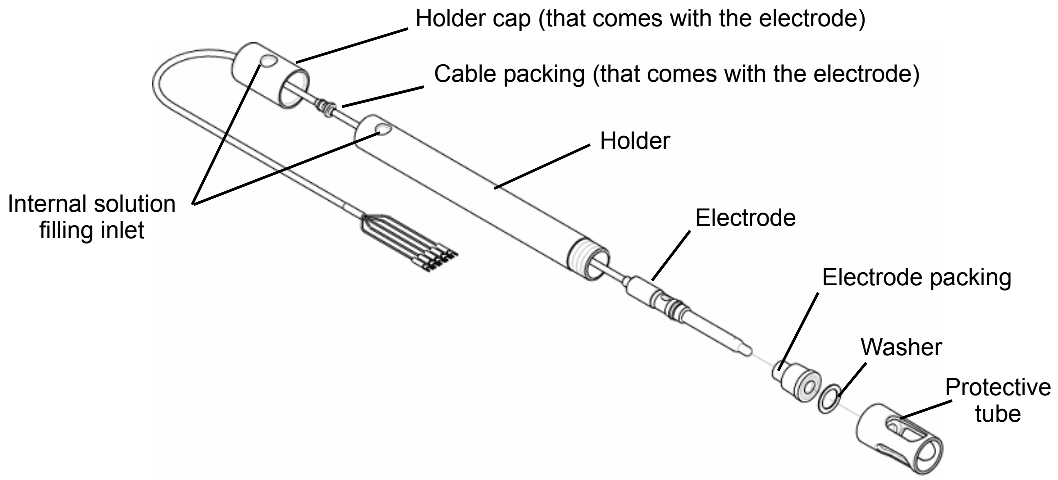


Storage

Ensure that the tip (glass membrane) and the liquid junction of the electrode are not dried. Fill the provided protective cap with tap water and then put the cap on the electrode tip for storage. In addition, tightly close the internal solution filling inlet to prevent internal drying.

Replacing the electrode

⚠ CAUTION
The electrode is made of glass and broken when exposed to impact or strong force. In handling the electrode, exercise sufficient care.

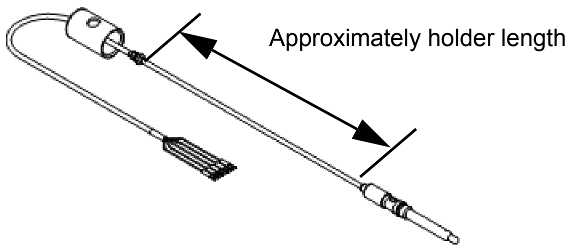


1. **Remove the holder cap and drain all the internal solution in the holder.**
2. **Remove the protective tube and the washer at the holder tip and then remove the electrode packing from the holder.**
3. **Remove the electrode packing while holding the electrode.**
4. **Remove the electrode from the top of the holder.**

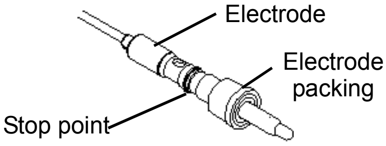
Note

Dispose of the pH electrode as industrial waste.

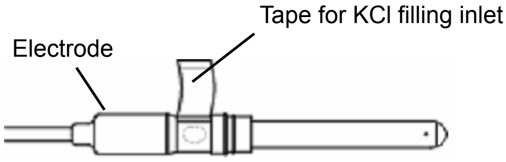
5. **Clean the holder, electrode packing, washer, and protective tube with alcohol or the like and then well dry them.**
6. **Extend the holder cap and the cable packing approximately to the holder length.**



7. **Insert a new electrode from the top of the holder and then pass it through the bottom of the holder.**
8. **Fully fit the electrode packing to the electrode.**



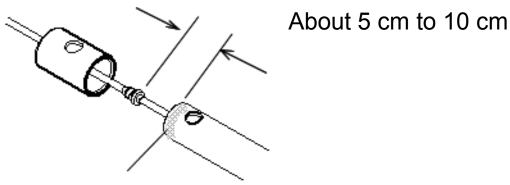
9. **Remove the protective cap for the electrode and the tape for the KCl filling inlet on the electrode.**



Note

- **Unless the tape for the filling inlet on the electrode is removed, no measurements can be made successfully.**
- **Since the internal solution in the electrode may leak, remove the tape while orienting the KCl filling inlet upward.**
- **Do not discard the protective cap as it is used for storage.**

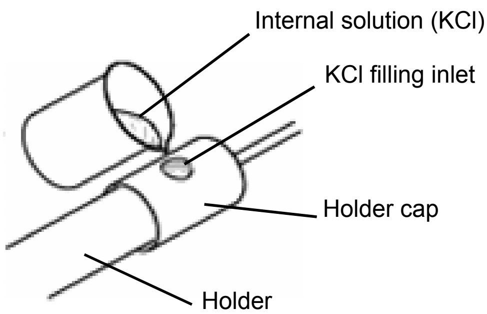
10. **Push the electrode packing into the holder and then tighten the protective tube through the washer.**
11. **Adjust the cable packing so that the distance between the holder cap and the holder top becomes about 5 to 10 cm.**



12. **Apply silicone grease to the entire circumference of the holder top.**
13. **Supply a new internal solution through the filling inlet up to the specified level.**

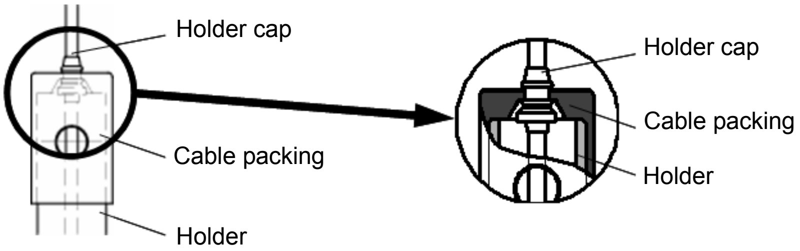
Tip

Approximately 500 mL of internal solution is required for the holder of 1 meter.

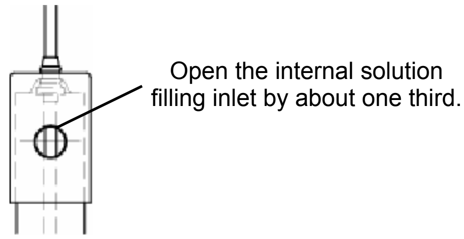


14. **Fit the holder cap into the holder.**

15. **Slightly draw the cable packing out of the holder packing.**



16. **Open the internal solution filling inlet by about one third.Now the electrode is ready for use.**



Note

Check the following points:

- **Did you remove the protective cap for the electrode?**
- **Did you remove the tape for the KCl filling inlet on the electrode?**
- **Is the filling inlet on the holder opened by about one third?**

Troubleshooting

Troubles with measurement

Symptom	Possible cause	Remedial action
Fluctuation in measured value	● Air bubbles adhere to the electrode. ● Air bubbles are generated due to a fast flow rate. ● The level of solution under measurement fluctuates, causing the wetted area of the electrode to vary.	● Ensure that no air bubbles are generated in the solution under measurement. ● Control the flow rate. ● Install the HP-200 so that the level of solution under measurement does not fluctuate.
	● A sudden change in solution temperature has occurred. ● A screw on the terminal block is loosened. ● The insulation of the terminal block deteriorates.	● Install the HP-200 in a location free from sudden changes in solution temperature. ● Firmly tighten the screw. ● Remove moisture from the terminal block ● Clean the terminal block.
Abnormal reading	● The electrode is dirty. ● The electrode is not immersed in the solution under measurement.	● Clean the electrode. ● Install the HP-200 so that the level of solution under measurement does not fluctuate.
	● The electrode cable is broken or short-circuited. ● A contact failure occurs on the terminal block.	● Check the electrode cable. ● Check the terminal block.
Slow response	● The flow rate is low. ● The solution under measurement has uneven concentrations (when a tank was installed).	● Increase the flow rate. ● Sufficiently agitate the solution. ● Install the tank in a location where the concentration becomes even.
Abnormal operation	● The circuit board is corroded by corrosive gas.	● Send clean instrumentation air from the air purge inlet.
Failure to turn ON the power	● The power supply voltage falls outside the rated voltage range. ● There is a wiring mistake. ● The HP-200 has an internal fault.	● Check the power supply voltage. ● Check the power wiring. ● Contact us.

Measurements exceeding the measurable range

When the measured pH value falls outside the measurable range (between -0.01 and 14.01 both exclusive), it blinks.
Take remedial action as instructed below:

Symptom	Possible cause	Remedial action
Blinking display (out of the measurable range)	The electrode is not immersed in the solution under measurement.	Install the HP-200 so that the electrode remains immersed in the solution under measurement even if the level of that solution fluctuates.
	The protective cap remains put on the electrode.	Remove the protective cap.
	● The electrode cable is improperly wired. In particularly, there is electrical discontinuity in the G- or R-line. ● The relay cable is improperly wired.	Check for any loosened screws on the terminal block in the HP-200 and that in the relay box and for any wrong wiring.

Error codes

The HP-200 has a capability of displaying various error codes.
An error code is displayed on the auxiliary display part while blinking.

Description of error codes

Display		Priority levels of errors*1	Error	Description	Occurs during
E-11	SENSOR ERROR	9	Response speed error	The response of the pH electrode is low during calibration using a standard solution.	pH calibration*2
E-12		10	Electrode sensitivity error	The sensitivity of the pH electrode decreases during calibration using a standard solution.	
E-13		11	Asymmetry potential error	Electromotive force (asymmetry potential) for pH7 is large during calibration using a standard solution.	
E-14		12	Standard solution error	The specified standard solution is not used.	
E-15		13	Standard solution temperature error	The temperature is 55°C or higher at pH10.	
E-21		4	Temperature sensor disconnection error	The lead wire from the temperature sensor has electrical discontinuity.	Measurement or pH calibration*2*3
E-22		5	Temperature sensor short-circuit error	The lead wire from the temperature sensor is short-circuited.	
E-25		8	Temperature measurable range error	The temperature does not fall within a range between -5°Cand 105°C when temperature sensor type is set to "Auto." The temperature does not fall within a range between -20°C and 130°C when temperature sensor type is set individually.	
E-27		8	Temperature calibration error	The temperature does not fall within a range between -20°C and 130°C.	Temperature calibration
E-71		6	Glass membrane error	The response membrane of the pH electrode is broken.	Measurement
E-72		7	Comparison electrode error	The liquid junction of the comparison electrode is clogged (only applicable when the wetted pole exists).	
E-90	SYSTEM ERROR	1	System error	Internal communication error	Any time (system error)
E-91		2	System error	The stored data such as settings and calibration values has been lost.	
E-92		3	System error	The A/D meter malfunctions.	

*1: More than one error cannot be displayed at a time. If multiple errors occur, the error having the higher priority level (smaller number) is displayed.

*2: For E-11 to E-15, re-calibration can be performed. However, if any of E-21 to E-25 occurs during calibration, no re-calibration can be performed.
Take remedial action in accordance with " Remedial action for error codes " (page 72).

*3: If no temperature electrode is used (temperature sensor type is set to "non," no error is displayed.

For the relationship between errors, alarm output, FAIL signal output, cleaning output, and transmission output, see " Output conditions for various output types " (page 63).

Tip

When an error code is displayed, the HOLD indicator blinks and transmission output becomes the output value specified on the Setup menu. See " Set currently held value for transmission output 1 " (page 36) and " Set currently held value for transmission output 2 " (page 37)

Remedial action for error codes

When an error code is displayed, take remedial action in accordance with the table below:

<table>
<tr><th>Error code</th><th>Trigger condition</th><th>Reset condition</th><th>Possible cause</th><th>Remedial action</th></tr>
<tr><td rowspan="2">E-11
(response speed error)</td><td rowspan="2">The reading has not become stable for 5 minutes or more during calibration stability detection.</td><td rowspan="2">● Press the ESC key.
● Perform re-calibration.</td><td>● The electrode is dirty.
● The electrode has remained dry for a long period of time.</td><td>Clean the electrode.
When the glass membrane is dried, the response deteriorates. Immerse the electrode in deionized water for 24 hours and then perform calibration again using a standard solution.</td></tr>
<tr><td>The temperature difference between the solution under measurement and the standard solution is large.</td><td>Wait until the temperature of the temperature compensation element in the electrode becomes stable and then perform calibration using a standard solution.</td></tr>
<tr><td>E-12
(electrode sensitivity error)</td><td>The electrode sensitivity (SLOPE) does not fall within a certain range during calibration for more than one point.
The error occurs when SLOPE < 40 mV/pH or SLOPE > 65 mV/pH.</td><td>● Press the ESC key.
● Perform re-calibration.</td><td>● The electrode is dirty.
● The glass electrode is broken.
● The electrode's internal solution is abnormal.
● The internal solution level is low.</td><td rowspan="3">● Clean the electrode.
● Replace the electrode if broken.
● For the internal solution adding type, add the internal solution if its level is low.
● For the internal solution non-adding type, replace the electrode if no white powder is seen in the solution.
● If the internal solution shows discoloration or deterioration, change the whole internal solution.
● If you use an old standard solution, perform calibration using a new standard solution.
● Use an appropriate standard solution.</td></tr>
<tr><td>E-13
(asymmetry potential error)</td><td>The asymmetry potential (STD) does not fall within a certain range during calibration.
The error occurs when STD > 90 mV or STD < -90 mV.</td><td>● Press the ESC key.
● Perform re-calibration.</td><td>● The electrode is dirty.
● The internal solution shows discoloration.
● The internal solution is contaminated.
● The pH standard solution is abnormal or it deteriorates.</td></tr>
<tr><td>E-14
(standard solution error)</td><td>● During calibration for more than one points, the difference in pH value between two solutions is not 2.0 or more.
● During calibration for more than two points, the solutions for acidity, pH7, and alkalinity are not available
● The standard solutions used are not applicable for pH2, 4, 7, 9, and 10.</td><td>● Press the ESC key.
● Perform re-calibration.</td><td>● The difference in pH value is not enough (pH2 or less).
● The electrode is faulty.
● The pH standard solution is abnormal or it deteriorates in quality.</td></tr>
<tr><td>E-15
(standard solution temperature error)</td><td>The temperature during calibration using a standard solution for pH10 is 55°C or higher.</td><td>● Press the ESC key.
● Perform re-calibration.</td><td>The temperature of standard solution for pH10 is 55°C or lower.</td><td>Decrease the temperature of that solution.</td></tr>
</table>

Error code	Trigger condition	Reset condition	Possible cause	Remedial action
E-21 (temperature sensor disconnection error)	● When temperature sensor is set to 500 or 1k, its resistance value is approx. 1.58 kΩ or more (when the temperature sensor for 1 kΩ is approx. 150°C). ● When temperature sensor is set to 6.8k, 10k, or "Auto," its resistance value is approx. 14.8 kΩ (the temperature sensor for 10 kΩ is approx. 150°C).	The left condition no longer exists.	The resistance between T and T of the electrode is abnormal.	If the resistance value does not fall within the left range, the electrode is faulty. Replace the electrode.
			The electrode has no temperature sensor.	Set sensor type to "non." See“ Set sensor type ” (page 32).
			The electrode cable or the relay cable is improperly wired.	Check the wiring between T and T for being open.
E-22 (temperature sensor short-circuit error)	● When temperature sensor is set to 500, 1 k, or "Auto," the temperature sensor's resistance value is approximately 400 Ω or less (the temperature sensor for 500 Ω is at approximately -27°C or less). ● When temperature sensor is set to 6.8 k or 10 k, the temperature sensor's resistance value is 4.84 kΩ or less (the temperature sensor for 6.8 kΩ is at approximately -50°C).	The left condition no longer exists.	The resistance between T and T of the electrode is abnormal.	If the resistance value does not fall within the range specified on the left, the electrode is faulty. Replace the electrode.
			The electrode cable or the relay cable is not properly wired.	Check the wiring between T and T for short-circuit.
E-25 (temperature measurement range error)	● Temperature type: "Auto" The temperature does not fall within a range between -5°C and 105°C both exclusive. ● Temperature type: Individual The temperature does not fall within a range between -20°C and 130°C both exclusive.	The left condition no longer exists.	The temperature of solution under measurement is abnormal.	Ensure that the temperature of solution under measurement falls within the operating temperature range of the electrode.
			Temperature sensor is set to a wrong type.	Set temperature sensor to the correct type. See “ Set sensor type ” (page 32).
			The resistance of temperature electrode is abnormal.	The temperature electrode is faulty. Replace the electrode.

Error code	Trigger condition	Reset condition	Possible cause	Remedial action
E-27 (temperature calibration error)	The temperature does not fall within a range between -20°C and 130°Cboth exclusive.	You have exited the temp cal mode by pressing the ESC key.	The temperature of solution under measurement is abnormal.	Ensure that the temperature of solution under measurement falls within the operating temperature range of the electrode.
			Temperature sensor is set to a wrong type.	Set temperature sensor to the correct type. “ Set sensor type ” (page 32).
			The temperature electrode's resistance is abnormal.	The temperature electrode is faulty. Replace the electrode.
E-71 (glass membrane error)	The impedance of the pH electrode's response membrane and comparison electrode or wetted pole has become approximately 100 kΩ or less.	The left condition no longer exists.	The electrode is not immersed in the solution under measurement.	Immerse the electrode in the solution under measurement.
			The specified relay cable land relay box are not used.	Use the specified relay cable and relay box.
			The response membrane of the electrode has cracking.	Replace the electrode.
E-72 (comparison electrode error)	The impedance of the pH electrode's comparison electrode and wetted pole has become approximately 100 kΩ or less.	The left condition no longer exists.	The internal solution in the comparison electrode has run out.	Add the internal solution.
			The comparison electrode's liquid junction is clogged.	Clean the electrode. If this does not settle the problem, replace the electrode.
			The electrode has electrical discontinuity.	Replace the electrode.
			An electrode without a wetted terminal is used and no short-circuit plate is available.	The self-check capability is disabled. Set self-check is "non." “ Set sensor type ” (page 32).
E-90 (system error)	The internal communication in the HP-200 is abnormal.	Turn OFF the power and then turn it ON again.	The sensor connection is wrong.	Check the sensor connection and if it is wrong, connect the sensor properly. See “ Connecting the electrode cable ” (page 17).
			The internal system in the HP-200 is abnormal.	Turn OFF the power and then turn it ON again. If the system error does not disappear after the system has been restarted, contact us.
E-91 (system error)	The stored data such as settings and calibration values has been lost.		The internal system in the HP-200 is abnormal.	
E-92 (system error)	The A/D meter malfunctions.			

Remedial action for pH electrode's fault

If our pH electrode becomes faulty, take remedial action in accordance with the following table. If the pH electrode is not rectified after taking the remedial action, contact us.

Symptom / Possible reason	Calibration failure	Instable reading	Slow response	Unchanged reading	Fast consumption of internal solution	Foreign matter in comparison electrode	Short service life	Remedial action
Cracking of glass membrane/comparison electrode	○	○	-	○	○	○	-	Replace the electrode as it is unusable.
Dirty glass membrane	○	○	○	○	-	-	-	Clean the glass membrane with tap water or the like.
Dry glass membrane	○	○	○	-	-	-	-	Immerse the glass membrane in tap water for about 1 hour and then use the electrode.
Dirty or clogged liquid junction	○	○	-	○	-	-	-	Clean the liquid junction with tap water or the like.
Shortage of internal solution in comparison electrode	○	○	○	-	-	○	-	Add internal solution in comparison electrode.
10 mS/m (0.1 mS/cm) max. in electrical conductivity of solution under measurement	-	○	○	-	-	-	-	No measurements can be made. Contact us.
Unremoved cap	○	○	○	○	-	-	-	Remove the cap before use.
Closed internal solution filling inlet	○	○	-	-	-	○	-	Open the internal solution filling inlet.
Glass membrane damaged during cleaning	○	-	-	-	-	-	○	Use sponge or the like to clean the glass membrane so that it is not damaged.
Improperly connected terminal	○	○	-	-	-	-	-	Use our designated cable to connect the terminal in accordance with the instruction manual.
Lost liquid junction (ceramics)	-	-	-	-	○	○	-	Use the sleeve type or replace the electrode early.
Use of old standard solution	○	-	-	-	-	-	-	Use a new standard solution.
Hydrofluoric acid included in solution under measurement	○	-	○	○	-	-	○	Hydrofluoric acid melts glass. It is recommended that the electrode be replaced early.
Back flow of solution under measurement	○	○	-	-	-	○	○	Change the internal solution in the comparison electrode or replace the electrode.

Self-check capability for pH electrode

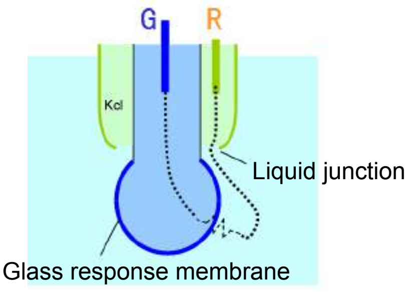
The HP-200 has a self-check capability for the pH electrode. This capability detects any cracking of the glass response membrane of the electrode or any clogging of the comparison electrode (liquid junction). It may not work depending on the electrode type or the operating environment. This section describes the self-check capability.

■ Self-check capability types

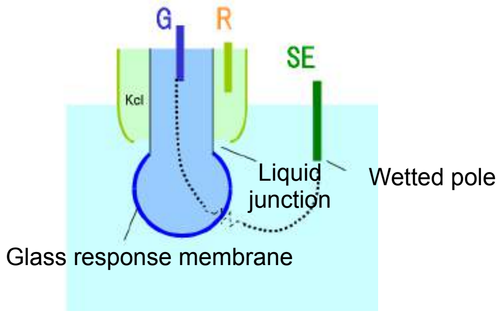
The self-check capability for the pH electrode is available in the following two types:

● Detection of cracking of glass response membrane (glass membrane error)

AC voltage is applied between the glass response membrane and the wetted pole or the comparison electrode to measure the impedance (resistance) between them.
When the measured resistance becomes below a certain threshold, the alarm for E-71 (response membrane error) is triggered.



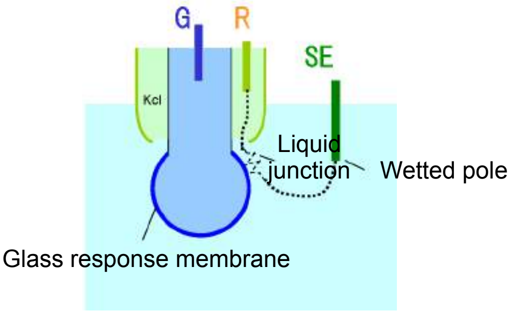
Electrode without wetted pole



Electrode with wetted pole

● Detection of liquid junction resistance error (comparison electrode error)

AC voltage is applied between the comparison electrode and the wetted pole to measure the impedance (resistance) between them. When the measured resistance becomes below a certain threshold, the alarm for E-72 (comparison electrode error) is triggered.



Electrode with wetted pole

■ Self-check capability for different pH electrode types

● pH electrode without wetted pole (e.g., 6110, 6108, 6109, 6151, 6152, 8200, or 8300)

For this electrode, only cracking of the glass response membrane ("g") can be detected.

● pH electrode with wetted pole (e.g., 6171, 6172, 6173, or 6174)

For this electrode, both cracking of the glass response membrane ("g") and the liquid junction resistance (r) error can be detected.

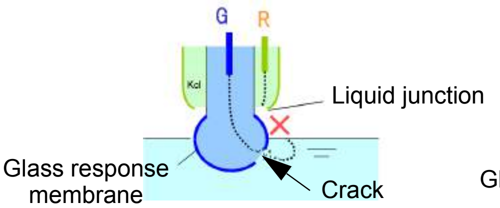
Cases where the self-check capability does not work properly.

The self-check capability may not properly work depending on the electrode type or the operating environment. This section describes the cases where this capability does not work properly.

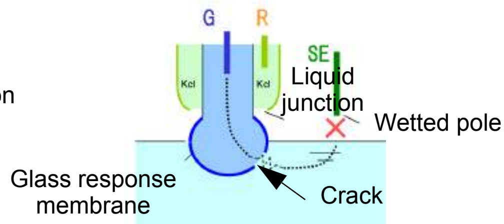
Case where the electrode is not wetted by the solution under measurement.

If the electrode does not come into contact with the sample, the self-check capability does not work properly.

- Even if the glass response membrane cracks, the response membrane error (E-71) does not occur.
- Even if the comparison electrode is normal, the comparison electrode error (E-72) occurs.



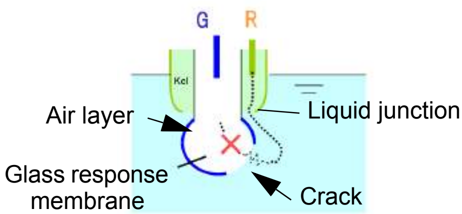
Electrode without wetted pole



Electrode with wetted pole

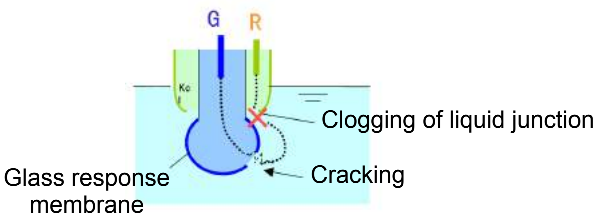
Case where an air layer exists in the glass response membrane

If there is an air layer in the glass response membrane when that membrane cracks, the resistance between poles cannot be successfully measured due to air insulation. In this case, therefore, the response membrane error (E-71) does not occur.



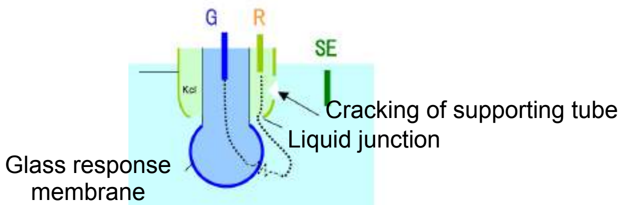
Case where the liquid junction has been clogged in an electrode without a wetted pole.

Even if the glass response membrane cracks when the liquid junction is clogged in an electrode without a wetted pole, the response membrane error (E-71) does not occur so far as the resistance of the liquid junction is large.



- **Case where the supporting tube for the comparison electrode cracks in an electrode with a wetted pole**

If the supporting tube for the comparison electrode cracks in an electrode with a wetted pole, the liquid junction resistance becomes smaller. In this case, therefore, the comparison electrode error (E-72) does not occur.



- **When the electric conductivity of solution under measurement is 10 mS/m (0.1 mS/cm) maximum:**

The self-check capability does not work properly because of its operating principle. In this case, disable this capability when using the HP-200.

- Even if the glass response membrane has cracking, the response membrane error (E-71) may not occur.
- Even if the comparison electrode is normal, the comparison electrode error (E-72) may occur.

Tip

To change the settings for the self-check capability, see " Set sensor self-check " (page 32).

Reference data

Specifications

Product name			pH meter
Model			HP-200
Combined sensor			Glass electrode
Measurement range	pH		pH0 to pH14 (display range: pH-1 to pH15)
	Temperature		0°C to 100°C When the automatic detection capability for temperature sensor types is used: Display range: -10°C to 110°C When temperature sensor type is manually set: Display range: -20°C to 130°C
Display resolution capability	pH		0.01 pH
	Temperature		0.1°C
Efficiency	pH	Repeatability	Within ±0.03 pH (in equivalent input)
		Linearity	Within ±0.03 pH (in equivalent input)
	Temperature	Repeatability	±0.3°C (in equivalent input)
		Linearity	±0.3°C (in equivalent input)
Transmission output	Number of outputs		2 (the negative terminals for transmission outputs are internally connected and at the same electric potential)
	Output type		4 mA to 20 mA DC; input/output isolated type
	Load resistance		Maximu:900 Ω
	Linearity		Within ±0.08 mA (output only)
	Repeatability		Within ±0.02 mA (output only)
	Output range	Output 1	pH: Freely selectable within a fixed range or the measurable range
		Output 2	Temperature: Freely selectable within a range between -20°C and 130°C
	Error output		With burnout capability (either 3.8 mA or 21 mA is selectable)
	Hold capability		Directly previous value, arbitrary value, or calibration value is selectable as the held value.
Contact output	Number of outputs		5
	Output type		No-voltage contact output
	Contact type		Relay contact; SPDT (1c)
	Output capacity		250 VAC 3 A, 30 VDC 3 A(resistance load)
	Contact function	R1, R2	Selectable from upper limit alarm, lower limit alarm, ON/OFF control, and time-sharing proportional control (The contact is closed during alarm operation, opened normally and while the power is down.)
		R3, R4	Selectable from upper limit alarm, lower limit alarm, ON/OFF control, currently holding of transmission output, and cleaning output (The contact is closed during alarm operation, opened normally and while the power is down.)
		FAIL	Error alarm (Closed in the normal state, opened in the failure state or while the power is down.)
	Description of error alarm		● Setting range:0.00 pH to 14.00 pH ● Delay time: 0 to 600 seconds

Contact output	Description of control	ON/OFF	● Setting range :0.00 pH to 14.00 pH ● Control width :0.02 pH to 4.00 pH (±0.01 pH to ±2.00 pH)
		Time-sharing proportion	● Setting range :0.00 to 14.00 pH ● Proportional band :0.02 pH to 4.00 pH ● Period: 5 to 300 seconds ● Control output shift function: 0% to 50% of the period can be shifted. ● Automatic period adjusting capability: When the deviation enters a certain range (F-zone), the period is automatically extended accordingly (this capability is disabled when the shift capability is enabled). ● F-zone: 1% to 100% of proportional band ● (When the deviation enters the above range, the automatic period adjusting capability is enabled.) ● Maximum period extending time: 0 to 300 seconds ● Maximum control quantity: 50% to 100% (regardless of the proportional band)
Cleaning output	Number of outputs		1
	Output type		Contact output with voltage (output of connected power supply voltage)
	Contact type		Relay contact; SPST (1a)
	Contact capacity		250 VAC 3 A, 30 VDC 3 A(resistance load)
	Contact function		Actuation of solenoid valve for cleaning
	Settings	Cleaning period	0.1 to 168.0 hours
		Cleaning time	2 to 600 seconds
		Hold time	2 to 600 seconds
	Timer accuracy		Within 2 minutes per month
	Description of cleaning operation		One can be selected from among the following. ● Operation of internal timer ● Operation of internal timer and external input ● Enable the internal timer only when external input is active. ● The cleaning start signals (when this signal is ON for 2 seconds minimum, the internal cleaning sequence starts).
Contact input	Number of inputs		1
	Contact type		No-voltage "a" contact for open collector
	Conditions		ON resistance: 100 Ω max. Open voltage: 24 VDC Short-circuit current: 12 mA DC
	Contact function		Choose external input for cleaning operation or transmission output holding input.
Communication capability	Communication method		RS-485
	Signal type		2-wire, input/output insulated type (not insulated from transmission output)
Temperature compensation	Applicable temperature element		Platinum resistor: 1 kΩ (0°C)
			Normal characteristic temperature-sensitive resistor :500 kΩ (25°C),6.8 kΩ (25°C),10 kΩ (25°C)
	Element selection method		Automatic temperature sensor type detection or manual selection (temperature compensation may be disabled)
	Temperature compensation range		0 °C to 100°C
	Temperature calibration capability		One-point calibration by comparison with reference thermometer

Calibration	Calibration method		Auto or basic calibration	
	Number of calibration points		Selectable from 1, 2, and 3 points	
	Kinds of standard solutions		pH 2, 4, 7, 9, and 10	
			Arbitrary standard solution (with difference of 2 pH min.) for basic calibration	
	Additional capabilities		Automatic detection of kind of standard solution	
			Automatic detection of electric potential stability	
			Automatic detection of calibration failure (asymmetry potential, sensitivity, or response time)	
			Calibration history (asymmetry potential, sensitivity, and number of days elapsed after last calibration)	
Self-check	Calibration error		Asymmetry potential error, sensitivity error, and response time error	
			Temperature calibration error	
			Standard solution detection error	
	Electrode diagnostic error		Cracking of glass response membrane	
			Comparison electrode impedance error (only applicable for electrode with liquid junction pole)	
			Temperature sensor short-circuit, temperature sensor disconnection, and deviation from temperature measurement range	
	Meter error		CPU error, ADC error, and memory error	
Operating temperature range			-20°C to 55°C (without freeze)	
Operating humidity range			Relative humidity: 5% to 90% (without condensation)	
Storage temperature			–25°C to 65°C	
Power supply	Power supply voltage range		90 V to 264 VAC 50/60 Hz	
	Power consumption		15 VA (max)	
	Others		With time lag fuse (250 V, 1 A)	
			With power switch for maintenance use	
Compatible standards	CE marking	Compatible standards	EMC Directive (2004/108/EC) EN61326-1:2006 Low-voltage Directive (2006/95/EC) EN61010-1:2001	
		Immunity	Industrial location	
			Electrostatic discharge	IEC61000-4-2
			Electromagnetic field of radiated radio frequency	IEC61000-4-3 (*Note 1)
			Electric fast transient/burst	IEC61000-4-4
			Surge	IEC61000-4-5 (*Note 2)
			Conducted interference induced by radio frequency	IEC61000-4-6 (*Note 1)
			Voltage dip, short-time power outage, and voltage change	IEC61000-4-11
		Emission	ClassA	
			Radiated disturbance	CICPR 11 CLASS A
			Noise terminal voltage	CISPR 11 CLASS A
		Low voltage	Contamination level 2	
	FCC Rules		Part15 CLASS A	

Structure	Installation	Outdoor installation type	
	Installation method	Mounted on 50A pole or wall	
	Protection code	IP65	IEC60529, JIS C0920
	Case material	Aluminum alloy (coated with epoxy-denatured melamine resin)	
	Material of fittings	SUS304	
	Material of cover	SUS 304 stainless steel (coated with epoxy-denatured melamine resin)	
	Material of display win-dow)	Polycarbonate	
	Display element	Reflective monochrome LCD	
External dimensions		180 (W) x 155 (H) x 115 (D) mm (excluding the fittings)	
Mass		Main body: Approx. 3.5 kg; cover and fittings: Approx. 1 kg	

Note

- **Note 1: The standard for effect on the reading by the electromagnetic field of the radiated radio frequency and by the conducted interference is within the measured pH value ± 0.25 pH.**
- **Note 2: When the sensor cable, the transmission cable, or the contact input cable is extended by 30 m or more, the surge test under the EMC Directive for CE marking is not applied.**
- **Note 3: An arrester (sparkover voltage: 400 V) is implemented for transmission output, contact input, and communication. However, use a most suitable surge absorption element on the connection lines in accordance with the ambient environment, the situation of equipment installed, and the externally connected equipment.**

Information on standard solutions

The HP-200 allows you to perform calibration using five kinds of standard solutions.
The pH values of pH standard solutions at each temperature are shown in the table below:

Temperature (°C)	pH2 standard solution (oxalate)	pH4 standard solution (phthalate)	pH7 standard solution (neutral phosphate)	pH9 standard solution (borate)	pH10 standard solution (carbonate)
0	1.67	4.00	6.98	9.46	10.32
5	1.67	4.00	6.95	9.40	10.24
10	1.67	4.00	6.92	9.33	10.18
15	1.67	4.00	6.90	9.28	10.12
20	1.68	4.00	6.88	9.22	10.06
25	1.68	4.01	6.86	9.18	10.01
30	1.68	4.02	6.85	9.14	9.97
35	1.69	4.02	6.84	9.10	9.92
40	1.69	4.04	6.84	9.07	9.89
45	1.70	4.05	6.83	9.04	9.86
50	1.71	4.06	6.83	9.01	9.83
55	1.72	4.08	6.83	8.98	-
60	1.72	4.09	6.84	8.96	-
70	1.74	4.13	6.84	8.92	-
80	1.77	4.16	6.86	8.88	-
90	1.79	4.20	6.88	8.85	-
95	1.81	4.23	6.89	8.83	-

List of parts

Optionally available parts

Part name	Model	Description
Holder	CH-101 series	Immersed type holder
	HIBX series	Immersed type holder for tip replaceable electrode (617X series)
	CF-301 series	Distribution type holder (pressurization type)
	CF-401S	Distribution type holder (pressurization type/SUS stainless steel)
	CF-501	Distribution type holder for tip replaceable electrode (617X series)
Cleaner	CH/UCF series	Ultrasonic cleaner (immersed/distribution type)
	JCH/JCF series	Water (air) jet cleaner (immersed/distribution type)
	BCH series	Brush cleaner (immersed type)
	CCH series	Chemical agent cleaner (immersed type)
	CBCH series	Chemical agent cleaner (immersed type) with brush

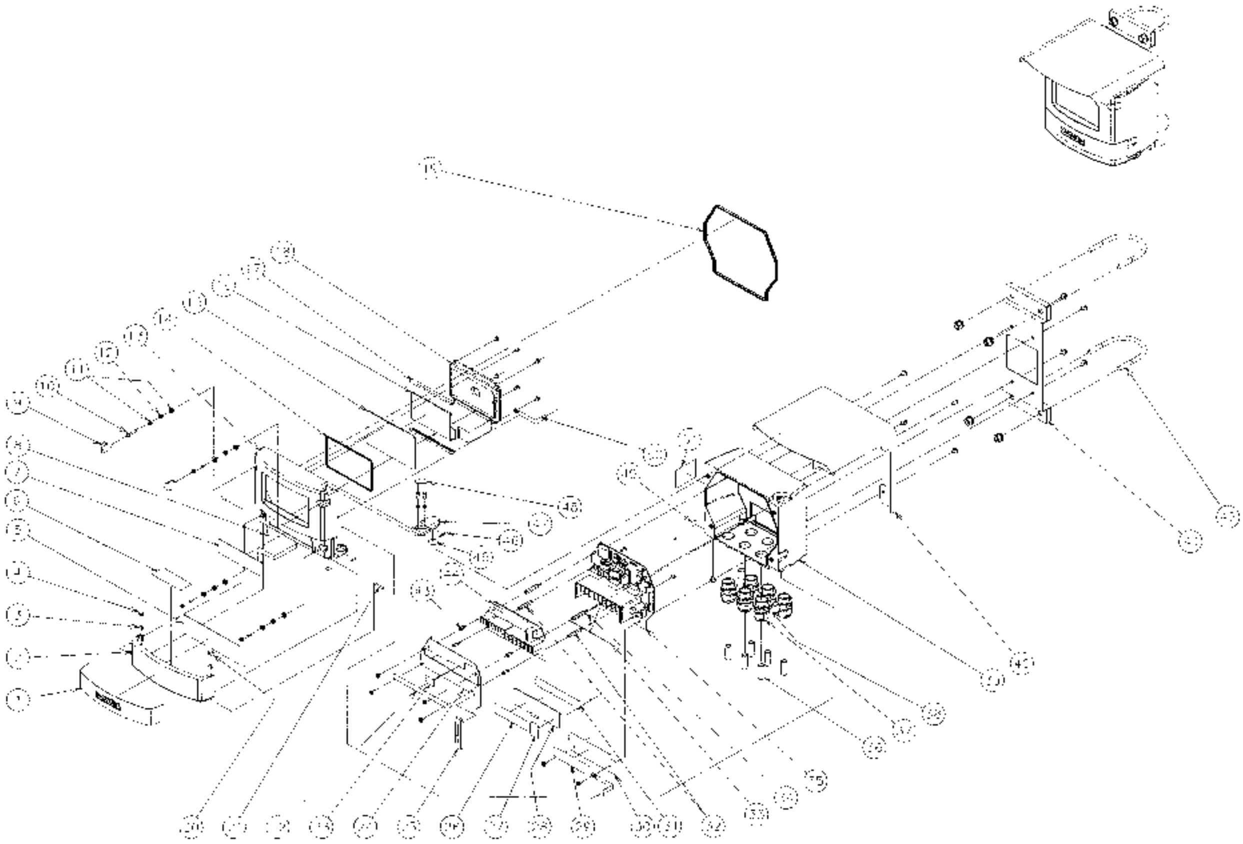
Spare parts

Part name	Model	Description		Part number
pH standard solution pH standard solution pH standard solution	#100-4 #100-7 #100-9	Standard solution for pH4 (accuracy: ±0.02 pH) Standard solution for pH7 (accuracy: ±0.02 pH) Standard solution for pH9 (accuracy: ±0.02 pH)	(500 mL) (500 mL) (500 mL)	3200043638 3200043637 3200043636
pH standard powder pH standard powder pH standard powder	#150-4 #150-7 #150-9	Standard solution for pH4 (accuracy: ±0.05 pH) Standard solution for pH7 (accuracy: ±0.05 pH) Standard solution for pH9 (accuracy: ±0.05 pH)	(containing 10 bags) (containing 10 bags) (containing 10 bags)	3200043619 3200043620 3200043621
Internal solution in comparison electrode	#300	3.33 mol/L KCl solution	(250 mL) Note: Two pieces (500 mL) are required for the 1 m holder.	3200043618
Powder for internal solution in comparison electrode	#350	KCl powder: 500 g	(1 piece)	3200043623

Part name	Model	Description				
		Type	Temperature compensation element	Operating temperature	Liquid-junction structure	Length of lead wire
pH composite electrode	6110-50B	General-purpose	Pt 1 kΩ (0°C)	0 °C to 60°C	Ceramic junction	5 m
	6108-50B	Tough		-10 °Cto 100°C		
	6109-50B	Touch sleeve		-10°C to 80°C	Sleeve	
	6151-50B	For hydrofluoric acid resistance		-10°C to 60°C	Ceramic junction	
	6152-50B	For high-alkaline resistance				
	6171-50B	Tip replaceable type for hydrofluoric acid resistance				
	6172-50B	Tip replaceable type for high-alkaline resistance				
	6173-50B	Tip replaceable type for oil resistance				
	6174-50B	Tip replaceable general-purpose type				
	8200	KCl-free throw-in type	None	0°C to 50°C		
	8300	KCl-free throw-in type	6.8 kΩ(25°C)			

Disposal method

To dispose of the HP-200, comply with the ordinances and regulations specified in your region. The parts for the HP-200 are made of the following raw materials:



No.	Name	Raw material	No.	Name	Raw material
1	Nameplate	PVC	27	Switch information plate	PET
2	Front cover	ADC12	28	Terminal cover L	PC
3	Press-fit plunger	POMÅASUS631	29	Warning plate	PE
4	Plunger packing L	Q	30	Circuit board cover 4C	PC
5	Operation information plate	PET	31	Base for terminal information plate	PET
6	Warning plate	PET	32	Spacer	C3604BD
7	Key sheet	PET	33	Terminal information plate	PET
8	Protective sheet	PVC	34	Spacer	C3604BD
9	Screw cap	Q	35	Printed circuit board (H4W-MTH-0x)	Printed circuit board
10	Screw	SUS304	36	Seal pin	PVC
11	Spring cap	SUS304	37	Cable ground	66 Nylon, EPDM
12	Compression spring	SUS304WPB	38	Plug	SUS304
13	Top case	ADC12	39	Bottom case	ADC12
14	Window plate packing	Q	40	Roof	SUS304
15	Transparent panel	PET	41	Ball bracket	SUS304
16	LCD	Glass	42	U-shaped bolt	SUS304
17	LCD packing	Q	43	Spacer	C3604BD

No.	Name	Raw material	No.	Name	Raw material
18	LCD holder	ADC12	44	Printed circuit board (H4W-ANL-0x)	Printed circuit board
19	Case packing	Q	45	PTFE plate	PTFE
20	Plunger packing R	Q	46	Protective sheet U	PVC
21	Parallel pin	SUS304	47	Upper hinge	ADC12
22	Terminal information plate	PET	48	Screw cap 2	Q
23	Terminal block information plate	PET	49	Functional ground information plate	PET
24	Terminal cover 4U	PC	50	FG mesh	
25	Cable clamp	Q	51	Rating plate	PET
26	Warning plate	PET			

● Separately disposing of printed circuit boards

When the internal printed circuit boards must be separately disposed of, perform the following steps:

Tool to be supplied

- Phillips screwdriver

Disassembling procedure

⚠ WARNING

Electric shock hazard
Be sure to check that no electric power is supplied before starting the procedure.

1. **Check that no electric power is supplied.**
2. **Remove all the wires from the terminals connected to printed circuit boards 35 and 44.**
3. **Remove four screws that are used to secure terminal cover 24.**
4. **Remove four pieces of spacer 43.**
   Printed circuit board 44 will be exposed in the assembled state.
5. **Remove one flat cable connected to a connector on printed circuit board 44.**
6. **Remove four pieces of spacer 32.**
7. **Remove two screws that are used to secure terminal cover 24.**
   Printed circuit board 35 will be exposed in the assembled state.
8. **Remove two flat cables connected to connectors.**

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